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# Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

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**ABSTRACT** After a load is restored or a line is reclosed, the machines keep moving, and the residual response can resemble a new intervention. We formulate load/line attribution from eight PMUs in the IEEE 39-bus system as change detection and isolation with switched differential-algebraic (DAE) post-change models. A bank of linearized DAE predictors compares intervention and recovery histories that start from estimated past states, integrates load severity in closed form, and keeps a restored intervention available as the cause of the continuing response. The same operators give Willsky–Jones signatures, from which we compute, before any experiment, each source's detection and isolation information, its minimum detectable change versus noise, and the spurious evidence created by forgetting the physical state at restoration. The analysis predicts detection at the first post-onset report and reset evidence at least two orders of magnitude above the alarm threshold. In an independent simulation evaluation (18 physical parents, 108 interventions, 648 paired replays) the bank attributes 644/648 interventions before restoration with no alarm in 11.8 h of normal records; resetting the state lowers timely attribution from 108/108 to 75/108. Re-run with unchanged code, the bank tolerates a twentyfold increase of the calibrated noise; independent per-PMU noise of 0.05% TVE suppresses most load attributions although the angles still carry at least 36 times the alarm evidence, which places the limit in the noise model rather than in the physics. The single-model bank processes reports faster than the 30-report/s input.

**INDEX TERMS** change detection and isolation, differential-algebraic equations, event attribution, phasor measurement units, physics-informed inference, power system dynamics.

## I. INTRODUCTION

AFTER a load is restored or a line is reclosed, the machines continue moving. A second intervention may begin before that response has decayed. With only eight PMUs in a 39-bus network, a change in a measured signal may therefore indicate a new source, recovery from an earlier source, or an imperfect physical prediction. The task is to distinguish these explanations using only the reports available at the decision time.

Physical restoration is not electrical recovery. A detector that returns to a flat reference after OFF can issue another alarm for the remaining oscillation, while suppressing alarms during recovery can hide a real intervention. In statistical terms, the post-change law of a second intervention depends on the history of the first. Classical change detection and isolation treats a single change of an otherwise stationary law [1]–[3], and the generalized-likelihood approach of Willsky and Jones detects jumps in a fixed linear system [4]. Power-system outage detectors compare network hypotheses under quasi-static or linearized models [5]–[7], and trajectory libraries diagnose faults from sparse PMUs [8]. None of these formulations makes a restored intervention compete, as a physical cause, with later interventions in a switched DAE network.

We address this gap with three contributions.

1) *Recovery-aware attribution as switched-DAE change detection and isolation.* Candidate histories specify intervention, source, onset, severity interval and restoration time. Their predictors are reduced from the network DAE, start from estimated past states and keep the state through restoration. Severity is integrated in closed form, and separate active and cause weights separate the present intervention from the originating cause of the continuing response (Section IV).

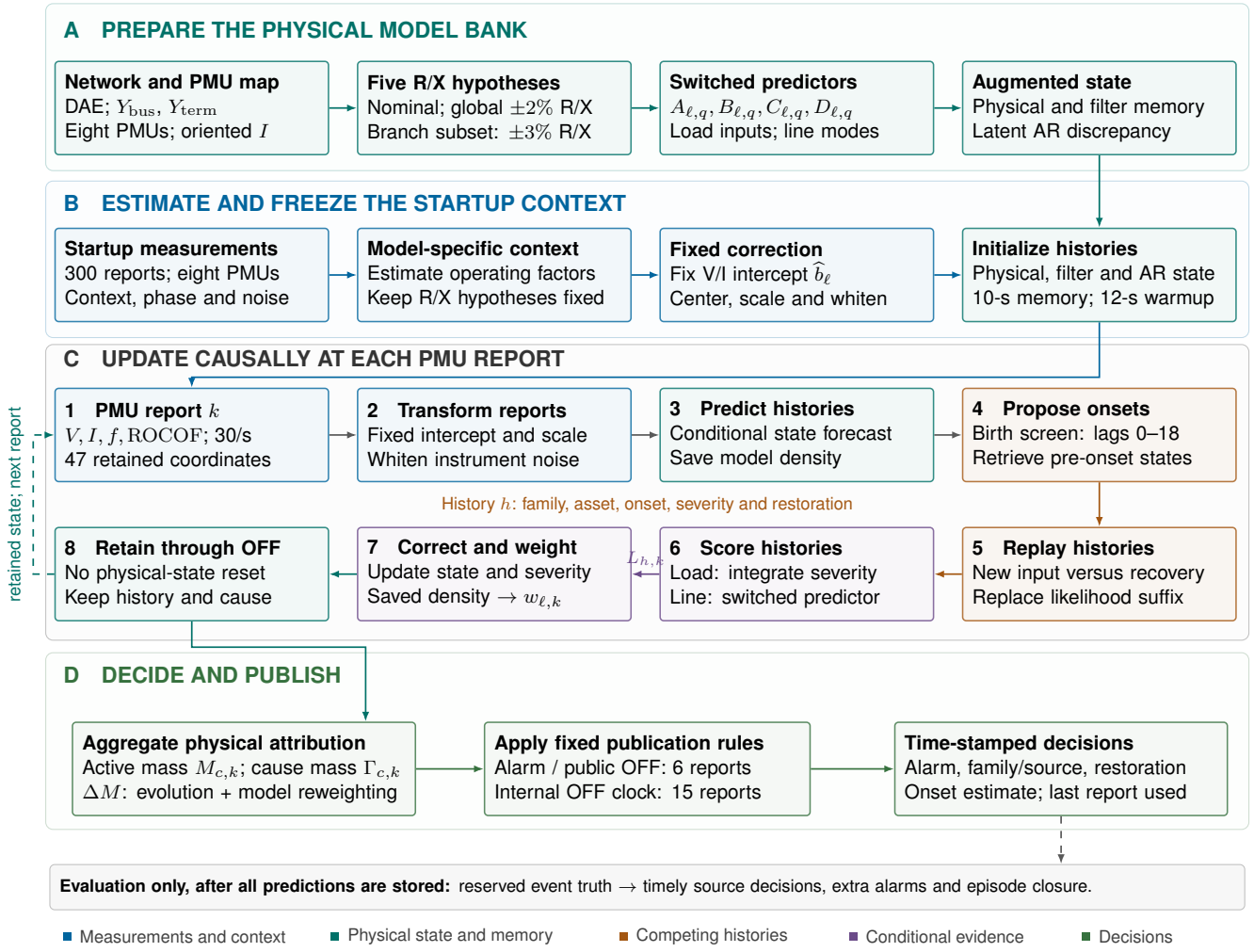



**FIGURE 1. Implemented physical-history attribution: model preparation (A), startup context (B), per-report inference (C) and publication (D). The bank predicts each report before correcting any state and stores the network-model evidence before proposals are selected. A replayed candidate replaces a likelihood suffix, so each report contributes once. The feedback loop retains physical and instrumental memory through restoration. Event truth is used only after all predictions are stored.**

2) *Physics-derived detectability and the cost of forgetting.* Because each history is linear-Gaussian given its mode path, its Willsky–Jones signature follows from the network operators. We compute the detection and isolation information of all 18 candidate sources, the minimum detectable load change as a function of PMU noise, and the spurious evidence created by resetting the physical state at restoration. These quantities are obtained before any campaign and are then checked against measured delays, a controlled state-reset study and a noise-stress test with unchanged code (Section V).
3) *An independent causal evaluation with explicit failure accounting.* On 18 new physical parents (108 interventions, 648 paired replays, 11.8 h of normal records) we separate detection, timely source attribution, unmatched alarms, restoration and computation time, and we report the published quickest-change detector of Chen *et al.* and two learned adapters at their original operating points (Section VII).

The bank attributes 644/648 interventions before physical restoration and raises no alarm on normal records. The analysis explains why: under the noise model calibrated on the reference data, a 10% load step or a line opening carries between about 300 and $5\times10^7$ times the evidence required for an alarm within one second. The errors that remain arise from network-model mismatch and from the publication rules, not from missing information. The same analysis identifies the conditions under which this ceases to hold. The supplement documents the implemented filter, the complete outcomes and the reproducibility contract.

## II. RELATED WORK

### A. CHANGE DETECTION AND ISOLATION IN DYNAMIC SYSTEMS

Sequential detection with a prior on the change time leads to the Shiryaev statistic, whose posterior-probability threshold

is optimal for a geometric change-time prior [9]. Detection *and isolation* among several post-change laws was formalized by Nikiforov [1] and Lai [2]; asymptotically, the delay for a given error level is inversely proportional to the Kullback–Leibler information separating the true post-change law from its competitors [3]. For linear state-space systems, Willsky and Jones introduced signatures of jumps propagated through a Kalman filter and estimated their magnitude by generalized likelihood ratios [4]. Separating a constant bias from the state, as Friedland did for recursive filtering [10], gives the affine state–severity representation used in our load histories. Interacting and multiple-hypothesis filters propagate mode-conditioned states and mode-sequence histories with pruning and merging [11], [12]. Our method inherits these mechanisms. Its post-change models, however, are switched DAE operators of a power network, and a restored intervention remains a competing cause of later observations.

### B. PHYSICAL INFERENCE OF GRID EVENTS

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [13]–[15]. Functional observability and descriptor-model diagnosability distinguish recovering a state quantity from identifying the input that produced it [16]–[19]. Phasor-angle hypotheses support outage identification [5], Bayesian outage inference with uncertain states [20], and quickest line-outage detection [6], [7]. Anguluri *et al.* recover sparse forcing locations and inputs of linear dynamics by group LASSO [21], and sensor placement determines which sources are distinguishable [22]–[24]. Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window [8] and derive error/delay guarantees for an adaptive matrix CuSum [25]. Trajectory sensitivities and Gaussian evidence have been combined for parameter inference in power-system DAEs [26], [27]. We use the same ingredients, DAE linearization, Gaussian evidence and onset search, to compute information-theoretic detectability of each catalog source and to test it against the attribution outcomes.

### C. LEARNED DIAGNOSIS FROM SPARSE MEASUREMENTS

Learned and knowledge-driven representations identify PMU disturbances and assets [28]–[30]. SHRED reconstructs IEEE-39 dynamic states from a few PMUs [31], and Dey *et al.* localize distribution topology changes with hardware-in-the-loop validation [32]. These methods learn the source map from labeled trajectories; the physical bank derives it from the network equations. We report two ExtraTrees adapters trained on the same 18-source catalog, whose limitations are discussed with the results.

## III. PROBLEM AND MEASUREMENT MODEL

At report $k$, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the reference record released with the SGSMA 2026 competition. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The causal stream is exported from one balanced complex phasor per PMU: phase A supplies the voltage and current magnitudes and angles. The CSV exporter constructs B and C by equal magnitudes and angle offsets of $-120^\circ$ and $+120^\circ$. Retaining A therefore loses no independent phase information in these synthetic records; it is not measured positive-sequence extraction or a model of unbalanced faults. The vector is $y_k = (|V_A|_{\mathcal{P}}, \angle V_{A\mathcal{P}}, |I_A|_{\mathcal{P}}, \angle I_{A\mathcal{P}\setminus\{39\}}, f_{\mathcal{P}}, \dot{f}_{\mathcal{P}})_k$. The supplementary measurement table gives all eight current-terminal assignments, polarities, gains and phase offsets. Products of voltage and current measured at different terminals are power proxies.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$\begin{aligned} h_k &= H_q(x_k, z_k, u_k), \\ \eta_k &= F_M(\eta_{k-1}, h_k; \theta_M), \\ y_k &= G_M(\eta_k, h_k; \theta_M) + b + \varepsilon_k. \end{aligned} \tag{3}$$

Here $x$ contains machine and control dynamics, $z$ the algebraic network variables, $q$ the topology/intervention mode, and $\theta_M$ the measurement convention. $H_q$ selects electrical phasors; $\eta_k$ retains the causal phase history used for frequency/ROCOF and the measurement-filter state. The observation thus depends on an augmented state. Bus frequency comes from phase evolution, not a machine's rotor speed. Instrumental noise is applied according to the declared generator contract.

**TABLE 1. Closest inference formulations. Only the quickest-change detector of Chen *et al.* and the two ExtraTrees adapters were executed on the common evaluation data.**

| Route | Mechanism | Evaluated target | Role in this study |
|---|---|---|---|
| Willsky–Jones GLR; Nikiforov/Lai | Filter signatures; detection–isolation statistics | One jump or one change in a fixed system | Information analysis (Section V) |
| IMM / M3H | Mode-conditioned filters, history pruning and merging | Hybrid switching-state estimation | History-bank mechanism |
| Chen / Rovatsos | Network likelihoods, sequential tests | Line outages | Chen executed on persistent openings |
| Warner high-fidelity | Simulated V/I paths, onset search | Short-circuit line candidates | Closest trajectory precedent |
| DAE / stochastic-state estimation | Physical state and parameter models | States, inputs or parameters | Components |
| Learned PMU diagnosis | Features or learned representations | Event families and locations | Two ExtraTrees adapters executed |
| This work | Switched-DAE histories with recovery | Successive catalog load/line events | Attribution, alarms, detectability |

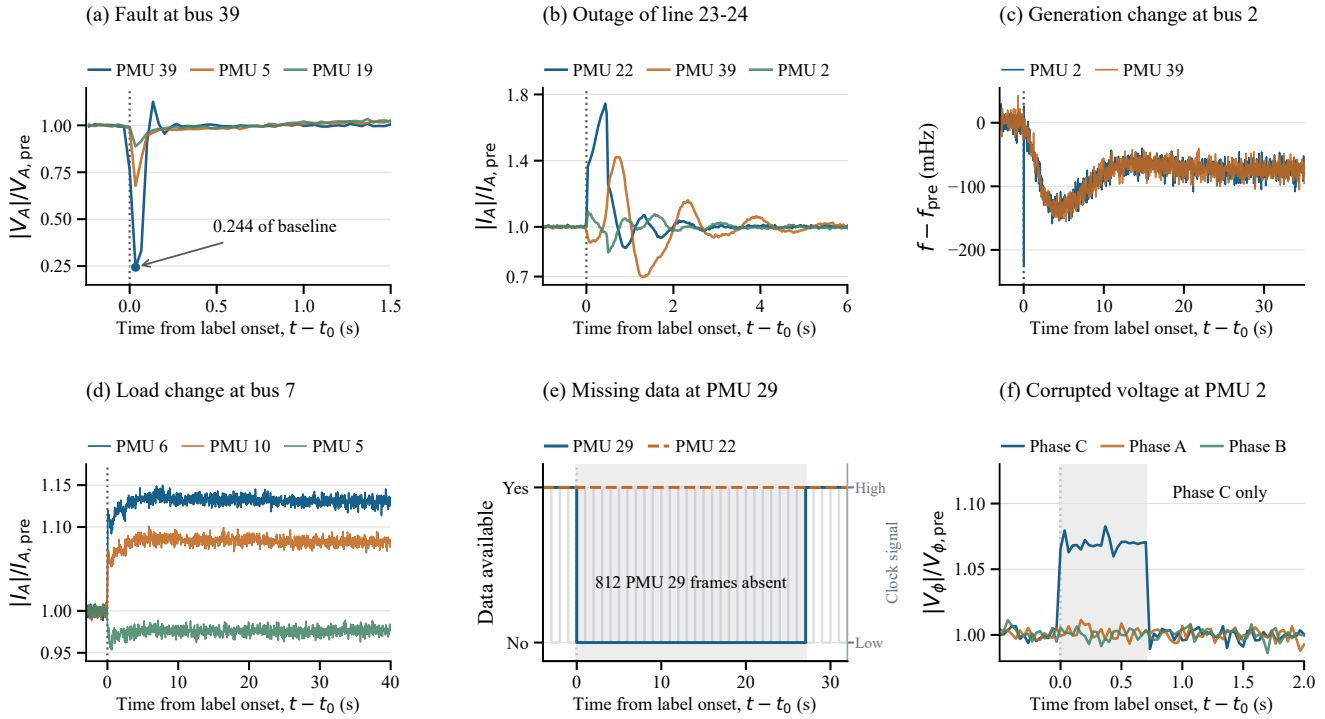



**FIGURE 2. Representative signatures from the archived RAW0001 reference simulation of the SGSMA 2026 competition: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Panels are aligned at the labeled onset for illustration only; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not a measured channel. Load and line mechanisms enter the causal evaluation; the other panels show the broader event taxonomy.**

An intervention history is the discrete tuple $h = (e, c, \tau, \kappa, \mathcal{I})$: family, source, onset, restoration time ($\kappa = \infty$ for a sustained event) and, for a load, a severity interval $\mathcal{I}$. The severity $a \in \mathcal{I}$ is a continuous latent variable conditioned on $h$ and integrated in the evidence; a line opening has no severity. A load changes its local P/Q setpoints by the common factor $1 + a$; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \quad (4)$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

Switch-aligned reports use the consistent post-switch algebraic solution. The local reduction requires an invertible algebraic Jacobian after fixing the angle reference; the catalog excludes islanding line alternatives. Successful solves and local residual checks validate the admitted cases, not global regularity for arbitrary switching topologies.

The generator stores event schedules and true states sepa-

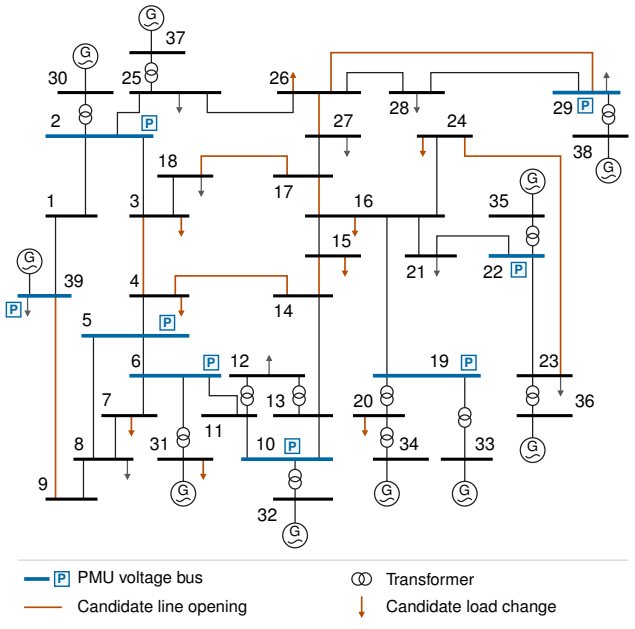



**FIGURE 3.** IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.

**TABLE 2.** Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

| Case | Physical operator | Executed claim |
|---|---|---|
| Normal | No intervention | Monitor after startup |
| Load | Local P/Q factor; optional restoration | Nine candidate buses |
| Line | Branch open; optional reclose | Nine candidate branches |
| Fault/generation | Different shunt/input/trip operators | Outside this causal test |
| Missing/bad data | Measurement-integrity mechanism | Outside the stream test |
| Unknown | Unresolved diagnosis | Not a new physical family |

rately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

## IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insufficient: competing histories must also yield a timely source decision.

### A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{\text{bus}} = Y_{\text{bus}} V, \quad I_{term,r} = Y_{term,r} V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

Each network hypothesis is linearized at the operating point $(\bar{x}_\ell, \bar{u}_\ell)$ determined by its prefix-fitted load and dispatch factors. For topology $q$, the constructor holds $\bar{x}_\ell$ fixed and solves $g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0$. On a regular index-one chart with a fixed angle reference, the derivatives at that consistent point give, suppressing $\ell$,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x \delta x + g_u \delta u), \\ \delta \dot{x} &= c_q + A_q^{red} \delta x + B_q^{red} \delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

Here $c_q = f_q(\bar{x}, \bar{z}_q, \bar{u})$ can be nonzero after a line opening even though the differential state is continuous. For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

The electrical offset is $d_q^{el} = H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u})$. Thus both the changed algebraic solution and the changed terminal map enter the prediction. These operators propagate an unmeasured-source perturbation to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in $g_q$ and $H_q$; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator also contributes directly through $H_q$, not only through the changed voltage. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

Exponential discretization integrates $c_q$ over each mode interval; composition handles a switch within a report. After instrumental augmentation, the resulting offsets enter $b_{\ell,q}$ and $d_{\ell,q}$ in

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q} a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q} a + v_k. \end{aligned} \tag{8}$$

Here $r_k$ contains reference-corrected, scaled and instrument-whitened PMU coordinates, $\ell$ is a fixed network hypothesis, and $\xi_k$ contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. $v_k$ denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture. The supplementary section "Implemented Predictor and Conditional Uncertainty" gives the affine construction and its code correspondence.

### B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

### C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual $e$, severity response $s$ and covariance $S$. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

With $\Phi$ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

As in separate-bias filtering [10], the conditional state mean is represented as $m_h + d_h a$; $d_h$ is the Willsky–Jones signature of a unit severity propagated through the filter [4]. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the one-report transition and observation operators for switch $o$. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With the shared covariance $P_\rho$ for the current discrepancy regime $\rho$,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \tag{13}$$

The amplitude distribution is updated by the same integral. Its truncated variance $V_{a,h}$ differs from the Gaussian parameter $v_h$. The affine conditional representation retains state–severity dependence:

$$\begin{aligned} \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ \mathrm{Var}(\xi \mid h) &= P_\rho + d_h V_{a,h} d_h^\top. \end{aligned} \tag{14}$$

Conditioning on past reports is implicit. The shared $P_\rho$ solves the nominal-operator Riccati equation of regime $\rho$; $S_o$ and $K_o$ are then formed for each switched operator $o$. This is a steady-state approximation, not an exact covariance recursion along each switched history. At closure, the filter retains the augmented-state mixture mean and records the amplitude/mean spread that it discards. The supplement separates these approximations from the exact local amplitude integral and states the covariance needed for a full mixture.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

### D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight is updated with the one-step predictive density

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}, \tag{15}$$

$$p_\ell(y_k \mid \mathcal{F}_{k-1}) = \sum_{h'} \pi_{\ell,h',k-1} \sum_{h \in \mathcal{T}(h')} p_\ell(h \mid h')\, L_{\ell,h,k}. \tag{16}$$

The transition set $\mathcal{T}(h')$ contains continuation, restoration at either half-report phase for an eligible active history, and a

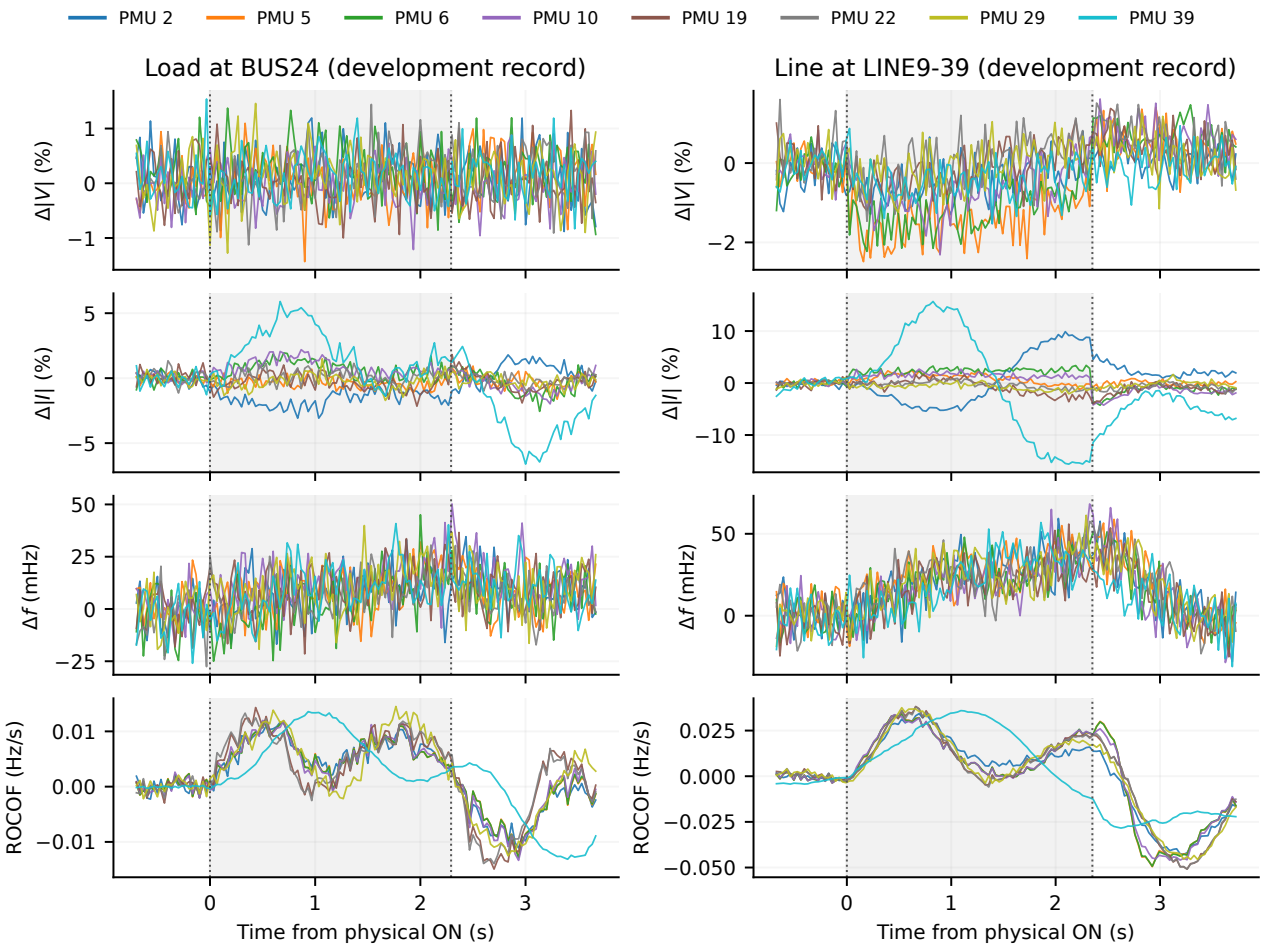



**FIGURE 4. Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.**

birth of every catalog cell and severity interval at either phase from the no-intervention history; the transition probabilities sum to one, which the implementation asserts at every report. The density is evaluated before any state is corrected and before retrospective proposals are selected. Onsets proposed later for earlier reports change the inner history posterior through suffix replay but do not re-weight $w_\ell$; this is the one point where the outer and inner recursions differ from a single joint posterior. Startup-fitting reports are not scored again. The mixture covers a finite set of network alternatives rather than a continuous impedance posterior.

### *E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION*

Restoring an input or topology at $\kappa$ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t,\kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t,\kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{17}$$

Here $y^{\mathrm{net}}$ denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a predicted trajectory.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell $c$:

$$\begin{aligned} M_{c,k} &= \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{18}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\Delta M_{c,k} = \sum_{\ell} w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) + \sum_{\ell}(w_{\ell,k} - w_{\ell,k-1})M_{\ell,c,k-1}. \quad (19)$$

The first term is within-model evolution and the second network-model reweighting. Because of pruning and the steady-state covariances, $M$ and $\Gamma$ are approximate attribution weights, not calibrated probabilities.

### F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

An alarm is published when the active mass $\sum_c M_{c,k}$ exceeds 0.99 for six consecutive reports; with a geometric change-time prior this is a Shiryaev-type posterior threshold [9], and the six-report confirmation suppresses brief posterior excursions. OFF is published when the restoration mass exceeds 0.99 for 15 reports. The diagnosis reports the family with the largest cause mass and its highest-mass source; episode-replacement rules handle a different source that begins before the previous episode closes. Every message records its emission time and the last report used, so a diagnosis can be correct but too late to precede physical OFF.

The nominal comparator is the same procedure with only the nominal network hypothesis.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1 - \exp[-(1/30)/30]$, fixed during method development before the independent schedules. They are deliberately conservative: test interventions last 2–5 s and are separated by 3–8 s, much shorter than the prior means of 60 and 30 s. A low birth hazard raises the evidence needed for an alarm by $-\log \pi_c$; Section V shows that this margin is small compared with the available information. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. Each interval is a separate history with its own weight and truncated amplitude distribution; its birth weight is the prior probability of that interval. The physical operators are shared across these four histories. For a fixed path with all four intervals retained, their weighted evidence equals the single-interval integral. This partition is a bank representation, not numerical quadrature or a piecewise physical approximation. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The supplement maps each operation to its implementation and lists the complete publication configuration.

### G. RELATION TO MULTIPLE-HYPOTHESIS FILTERING

For a retained history $h$, let $p_\ell(h \mid h')$ contain the admitted birth, continuation and restoration transitions. Before pruning, its normalized evidence update has the standard form

$$\pi_{\ell,h,k} \propto L_{\ell,h,k} \sum_{h'} p_\ell(h \mid h')\pi_{\ell,h',k-1}. \quad (20)$$

This is the mode-sequence update of IMM and M3H filtering [11], [12]. Three features are specific to the present problem: the modes are oriented electrical intervention operators derived from the network DAE, histories carry a restoration time so that a restored intervention remains a competing cause, and the output separates active intervention from originating cause through (18). Retrospective births reuse a stored pre-onset state, and confirmed closure compresses histories into the retained augmented-state mean.

### H. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

**Algorithm 1: causal physical-history attribution.** The inputs are the nominal model, the fixed method configuration and the successive PMU reports; event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize conditional states and retain the eligible past states needed by retrospective proposals.
2) At report $k$, apply the fixed centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report. Compute and save the outer model predictive density before current-report screening and state correction.
3) Apply the short-lag birth screen using retained past states. A proposed history starts before its selected onset; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score candidate histories and replayed suffixes before correcting their states with the reports being scored. Integrate the allowed load severity and apply the switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass $M$, cause mass $\Gamma$ and $\Delta M$. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. Decisions use only reports already received. Computation time is reported separately in Section VII.

## V. PHYSICS-DERIVED DETECTABILITY AND THE COST OF FORGETTING

Each retained history is linear-Gaussian given its mode path. The information that the eight PMUs carry about every candidate intervention can therefore be computed from the predictor operators before any attribution experiment is run. This section derives three such quantities and states what they predict; Sections VII and VII-F test the predictions.

### A. SIGNATURES AND INFORMATION

Let intervention $i$ start at report 0 with severity $a$ and let the plant follow the local switched model (8) without noise. Run the conditional filter of candidate history $j$, started at the same onset, on these reports. Its innovation $e_j(k)$ and severity sensitivity $s_j(k)$ follow the recursion (13). The filter of the true history at the true severity has zero innovation, so the expected log-likelihood ratio of $i$ against $j$ after $k$ reports is the whitened signature energy

$$\Lambda_{ij}(k) = \tfrac{1}{2} \min_{b\in[l,u]} \sum_{r=0}^{k} \left\| e_j(r) - s_j(r)\, b \right\|^2_{S_j^{-1}}, \qquad (21)$$

where the minimization over the competitor's severity is omitted when $j$ is a line. With $j$ the no-intervention history, $\Lambda_{i0}$ is the *detection information*; $\min_{j\neq i} \Lambda_{ij}$ is the *isolation information*. Both follow from the reduced input directions, the topology offsets $c_q$ and $d_q^{el}$, the terminal map and the instrument model; no data are fitted. For a load, $\Lambda_{i0} = \frac{1}{2} a^2 G_i(k)$, with $G_i$ the accumulated information per unit severity squared. In change detection–isolation theory the asymptotic delay at a fixed error level is inversely proportional to this information rate [2], [3].

With birth prior $\pi_c$ per report and cell, the active mass of the true history at its own onset reaches 0.99 once

$$\Lambda_{i0}(k) \;\geq\; \lambda^\star = \log 99 - \log \pi_c, \qquad (22)$$

which is $\lambda^\star \approx 15$ nats for the configuration of Section IV. The minimum load change detectable within a horizon $T$ under a noise standard-deviation multiplier $\alpha$ is then

$$a_{\min}(\alpha, T) = \alpha \sqrt{2\lambda^\star / G_i(T)}. \qquad (23)$$

### B. THE COST OF FORGETTING THE PHYSICAL STATE

Suppose the conditional mean tracks the augmented state until an inferred closure at report $k_c$. If the state is carried, the innovations after $k_c$ keep zero mean. If the physical coordinates are set to zero while the instrument and discrepancy states are kept, the no-intervention filter receives the deterministic bias

$$\bar{e}_0(k) = C_0 \Phi_0(k, k_c)\, \delta x(k_c) - \sum_{r=k_c}^{k-1} C_0 \Phi_0(k, r+1) K_0\, \bar{e}_0(r), \qquad (24)$$

where $\delta x(k_c)$ is the zeroed physical part of the augmented state, $\Phi_0(k,r) = T_0^{k-r}$ and $C_0 = J_0$ are the no-intervention operators of (13), and $K_0$ is its gain. This is the free recovery response (17) seen through a filter that tries to absorb it. Any birth hypothesis $j$ that explains part of this bias gains $\Gamma_j^{sp}(k) = \Lambda_0^{reset}(k) - \Lambda_j^{reset}(k)$ over the no-intervention history, and a spurious alarm becomes possible when $\max_{j,k} \Gamma_j^{sp}(k) \geq \lambda^\star$. With carry, $\Gamma_j^{sp} \equiv 0$. The bias, and therefore the cost of forgetting, grows with the energy of the slow electromechanical modes excited by the intervention and with how strongly the PMUs observe them.

### C. PREDICTIONS

We evaluated (21)–(24) with the nominal operators built by the evaluated implementation in its pre-alarm (quiet) covariance regime. A consistency check confirms that the true-history filter leaves a residual below $10^{-8}$ nats. Fig. 5 gives the results.

*Prediction 1: detection is not information-limited under the calibrated noise.* A 10% load step carries between 4603 and $6.9 \times 10^6$ nats in one second, and a line opening at least $4.9 \times 10^6$ nats, against $\lambda^\star \approx 15$. The nearest competitor is separated by a comparable margin, so detection and isolation should both occur at the first post-onset report and the alarm should be limited by the six-report confirmation, i.e. at about 0.20 s. Doubling the noise should not change the delay.

*Prediction 2: a single detectability limit.* Under a uniform noise multiplier, every load except bus 31 remains detectable at the smallest tested change (3%) up to a multiplier of 20; the worst of them needs 1.9% there. The bus-31 load is only 9.2 MW, and its 3% change becomes undetectable within one second beyond a multiplier of about 7.4.

*Prediction 3: forgetting is costly everywhere.* After every one of the 108 interventions of the controlled study, resetting the physical state gives some birth hypothesis at least $2.2 \times 10^3$ nats of spurious evidence, more than $10^2$ times $\lambda^\star$; with carry the spurious evidence is zero.

These predictions are deliberately coarse. They use a single linearization, steady-state covariances and the true onset, and they ignore the proposal screen and the publication rules. Their value is that they are computed from the network equations alone and can fail.

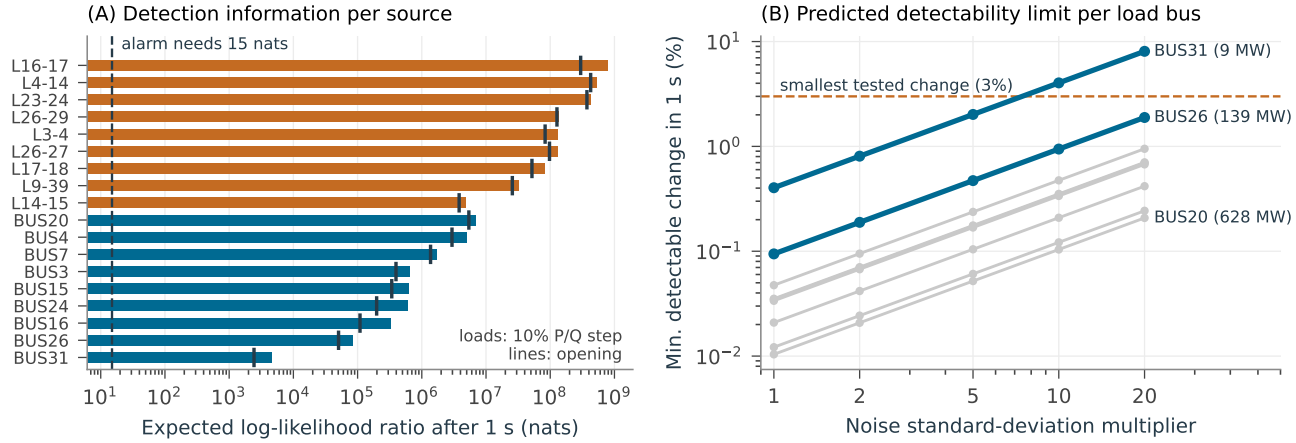



**FIGURE 5.** **Physics-derived detectability from the nominal switched-DAE operators, computed before the attribution campaign. (A) Expected log-likelihood ratio after 1 s against the no-intervention history (bars) and against the nearest competing source (vertical tick), for a 10% load step or a line opening; the dashed line marks the evidence $\lambda^\star$ required for an alarm. (B) Minimum load change detectable within 1 s, (23), versus a uniform multiplier of the calibrated noise; gray lines are the other seven load buses.**

## VI. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.

The plant is an IEEE-39 model in Julia/PowerDynamics with Sauer–Pai machines, IEEE Type I exciters, TGOV1 governors and constant-impedance ZIP loads. Each physical parent is integrated continuously, preserving differential states through ON/OFF. The PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001. At a switch-aligned report the export uses the post-switch algebraic value. The bus-31 candidate is the existing 9.2-MW/4.6-Mvar load at the generator bus that serves as slack only in the initial power flow; the dynamic model has no slack bus, so its load change is absorbed by machine dynamics like any other.

Instrument noise follows a stochastic model fitted to normal segments of the competition reference record: per-family vector autoregressions with a strong common factor across PMUs. Voltage-magnitude noise is about 0.4% of nominal, but the noise of angle *differences* between PMUs is only $10^{-4}$ to $10^{-2}$ degrees, because most angle noise is common to all units. This small differential noise is what makes megawatt-scale changes observable at remote PMUs (Section V); it is a property of the reference data, smaller than the idiosyncratic error expected from field PMUs [33]. Section VII-F therefore tests larger and independent per-PMU noise.

The two development campaigns each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

**TABLE 3.** **Candidate assets fixed during method development, before the independent schedules. This is a known catalog rather than a representative sample of all network assets.**

| Family | Candidate physical sources |
|---|---|
| Load buses | 3, 4, 7, 15, 16, 20, 24, 26, 31 |
| Lines | 3–4, 4–14, 9–39, 14–15, 16–17, 17–18, 23–24, 26–27, 26–29 |

The five-model bank and its nominal version were fixed before the independent schedules were drawn. Both Extra-Trees models were trained specifically for the causal load/line catalog, using 32 earlier physical parents for training and eight for development validation. Both training sets include normal operation and all 18 candidate sources. The reference uses 846 causal features from 1-, 5- and 10-s windows; the short-window variant adds 611 features from 3-, 8- and 15-report windows. Each model has 400 trees with the same predeclared hyperparameters. They inherit feature families from the SGSMA project, but their fitted weights are distinct from the original hackathon submission. These models and their adapters were fixed before the present independent schedules; no retraining uses the current evaluation records. All methods receive the same 30-fps observations. The supplementary reproduction table records the training lineage and model hashes.

The learned adapter publishes a change after six consecutive identical argmax labels; its tree votes are uncalibrated scores. In its own development validation the reference model already attributed no load before OFF and the short-window variant four, while lines were attributed well; both were nevertheless retained unchanged under a predeclared rule. They are therefore reported as fixed adapters at their original operating points, not as the best attainable learned baseline. A retrained learner with a threshold calibrated to a common alarm budget

**TABLE 4. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.**

| Population | Configuration | Main outcome | Interpretation |
|---|---|---|---|
| Continuous development | Three event parents, six interventions each; one normal parent; three network conditions | Timely family/source, extras and restoration | Open development evidence, not independent confirmation |
| Independent continuous evaluation | 18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales | Paired attribution, false alarms, delays and failures | Prespecified comparison of unchanged methods |
| Persistent-line evaluation | Nine lines, three parent repetitions; same network/noise strata | First stopping decision, source and censoring | Eligible comparison with published line-outage QCD |
| Supplementary signature test | 57 dictionary and 85 test trajectories; four loads; three common noise seeds | Nominal, reference correction and mixed-signature correction | Known ON/OFF and family; separate from causal accuracy |
| Supplementary recovery geometry | 38 trajectories; four recovery histories and 12 correlated source-pair comparisons | Separation with known history and one uncertain severity direction | Supplied prior trajectory, onset, family and pair; separate from stream accuracy |

is the subject of a separate study.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [6]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared research-plant adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all simulated instrumental effects. We preserve both matched-law implementation checks and model-mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

The independent campaign specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

**TABLE 5. Held-out continuous-stream decisions. Each family has 324 replays of 54 distinct interventions; extras are unmatched alarms on event records. The ExtraTrees adapters keep their original operating points.**

| Method | Load | Line | Extras |
|---|---|---|---|
| Physical bank | 322/324 | 322/324 | 72 |
| Nominal bank | 322/324 | 321/324 | 92 |
| ExtraTrees | 15/324 | 315/324 | 29 |
| ExtraTrees short | 30/324 | 322/324 | 16 |

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent results are followed by a development example and the controlled component study.

## VII. RESULTS AND CONTROLLED COMPARISONS

### A. INDEPENDENT EVALUATION: TIMELY ATTRIBUTION AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched alarms on event records, and its nominal counterpart 643/648 with 92. These are paired replays of 108 interventions in 18 physical parents, not 648 independent events. On the 72 normal records (42,336 s, 11.8 h of monitored time per method) neither bank raises an alarm, which bounds the normal false-alarm rate below 0.26 per hour at 95% confidence (Poisson). Every matched diagnosis emitted before OFF names the correct family and source; the timely misses are late or absent diagnoses. Table 5 and Fig. 6 give the family split.

The extra alarms concentrate where the network model is wrong: 68 of the 72 occur when the true R/X perturbation lies outside the five-model bank, 68 alarms in 1.08 h of out-of-bank monitored event-record time (about 63 per hour), against four in the other four strata together. Out-of-bank

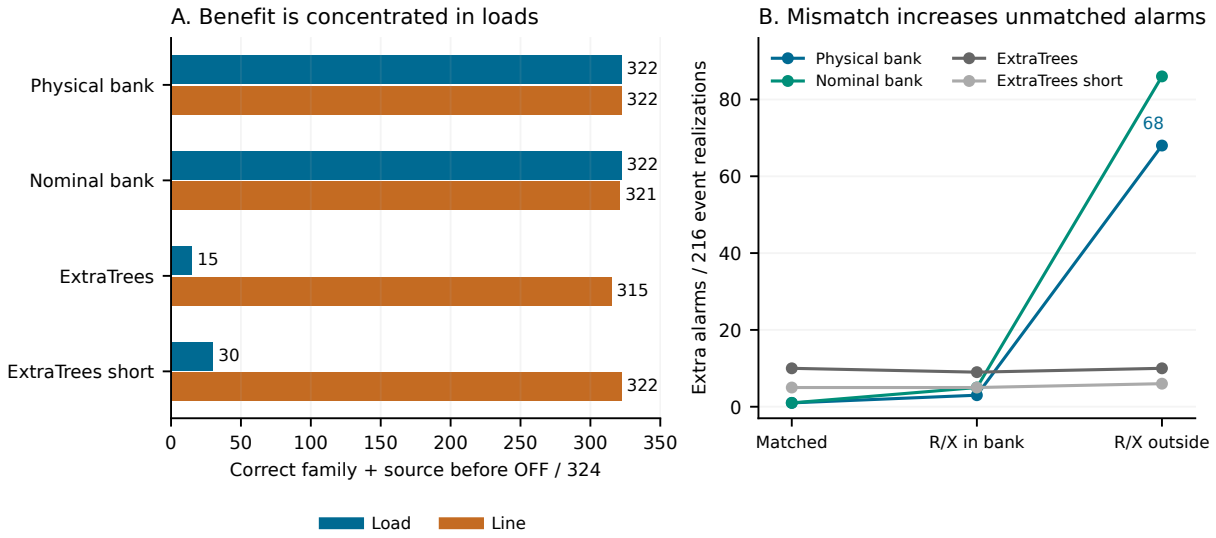



**FIGURE 6.** **Held-out continuous evaluation. (A) Timely correct family and source by event family, with 324 paired replays per family. (B) Unmatched alarms on event records for the two noise levels of each R/X condition (216 event replays per condition). Each method keeps its own publication rule.**

mismatch also produces the only four timely misses. Fig. 8 gives the paired parent-cluster intervals; the physical bank's median alarm delay is about 0.27 s in the matched and in-bank conditions, comparable to the learned adapters (0.19–0.30 s).

These delays test Prediction 1. For the nominal bank on matched networks, line alarms arrive after a median 0.18 s, at the six-report confirmation floor of 0.20 s less the sub-report onset phase, and doubling the noise leaves them unchanged. Load alarms arrive after 0.35 s, again unchanged by doubled noise but about 4 reports later than the available information requires. Line detection is therefore information-saturated, as predicted, while the load delay contains an algorithmic component that the information analysis exposes.

The two ExtraTrees adapters recover lines (315 and 322 of 324) but few loads (15 and 30). Fig. 7 shows that this is a coverage failure: every load diagnosis the adapters do emit before OFF is correct, but most load interventions receive none. A per-report audit on four further matched records (12 load and 12 line interventions, 1,245 and 1,143 in-event reports) locates the failure before any publication rule. During every load intervention the reference model's per-report argmax is the normal class in all reports, and the median of the largest true-class score is 0.09; during line interventions the true line wins 96% of the reports with median peak score 0.93. The short-window variant ranks the true load first in only 2 of 12 load interventions. The learned representation thus does not separate 3–14% load changes from normal operation, whereas the physical signatures carry hundreds to millions of nats per second for the same changes (Fig. 5). The same pattern was already present in their development validation (Section VI), so the comparison shows that the physical bank recovers load interventions that these particular adapters miss; it is not a test of the best attainable learned detector at a common alarm budget.

**TABLE 6.** **Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.**

| Stage | Median (s) | Range (s) | ms/report† |
|---|---|---|---|
| Bank build | 80.1 | 62.0–309.7 | 22.2 |
| History inference | 358.7 | 218.2–2641.1 | 99.6 |
| ExtraTrees (both) | 9.5 | 6.7–50.5 | 2.6 |
| Chen detector | 1.3 | 0.9–10.6 | 0.4 |

In the persistent-line test, both banks and both adapters identify the correct line first on all 162 openings. The adapter of Chen *et al.*'s quickest-change detector identifies 74/162; 66 records stop before the opening and 22 afterward on a wrong line, and it stops on all 72 paired normal records (first stop after 2,522 s of cumulative normal exposure, against 42,336 s monitored without alarm by the other methods). Its white-injection, quasi-static model does not describe the colored, dynamically filtered measurements of this plant, so its threshold is far from its nominal false-alarm rate. We therefore read this result as a test of transfer under model mismatch, not as a ranking of quickest-change detection. After the first identification, the two learned adapters raise 164 and 179 further unmatched alarms on the same persistent records, against 0 for the physical bank. Stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

### *B. COMPUTATIONAL COST*

Table 6 records the stages of the evaluation campaign, which ran the five-model bank on one thread. Its history inference

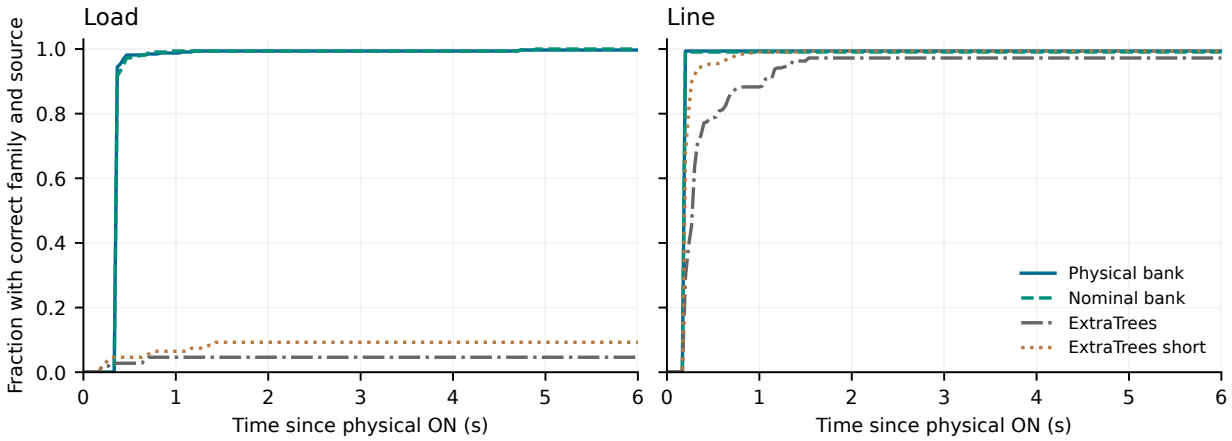



**FIGURE 7.** **Distribution of correct diagnosis time, using all 324 replays per true family in each denominator. An incorrect or absent diagnosis contributes no mass, and late correct decisions are included, so the curves differ from the before-OFF endpoint.**

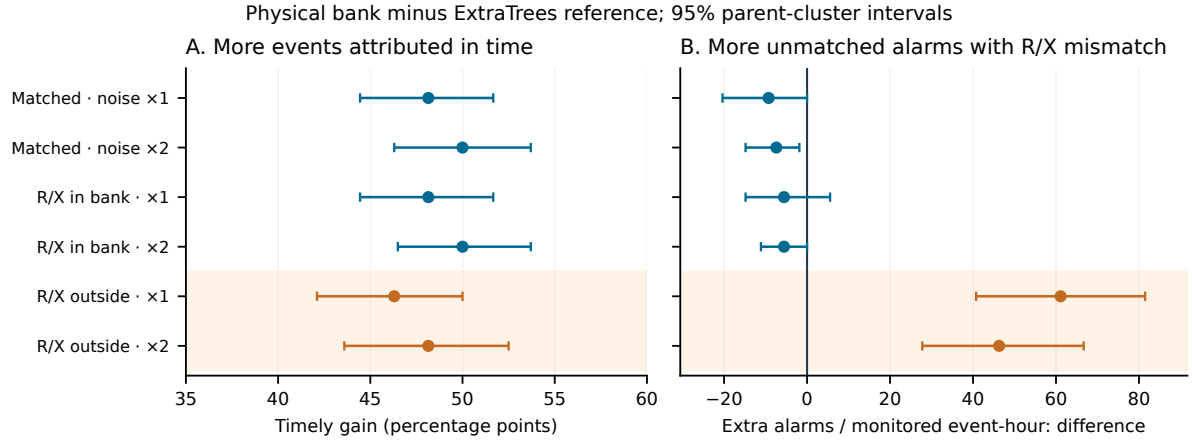



**FIGURE 8.** **Physical bank minus the ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per monitored event-record hour (the 12-s startup is excluded). Orange rows mark out-of-bank R/X conditions.**

takes a median 358.7 s per 120-s record (99.6 ms per report), three times slower than the 30-report/s input, so the five-model bank as implemented is an offline causal replay. The cost is dominated by the number of network hypotheses. Re-running the unchanged nominal bank alone on the same hardware and thread settings (Section VII-F) gives 26.4 ms per report (median over 6 records, maximum 30.0 ms), below the 33.3-ms report interval; the nominal bank attributed 643/648 interventions in the independent evaluation. These are amortized processing rates on a desktop CPU; per-report latency percentiles and memory were not instrumented, and the 80-s startup bank construction is a one-time cost. Decision timestamps elsewhere in the paper are report times, not wall-clock latency.

### C. A CONTINUOUS SEQUENCE AND A DELAYED DECISION

A development trajectory illustrates the timing criterion (Fig. 9). The network keeps moving after OFF, and a later event is evaluated against that predicted response. Under out-of-bank network error, the bus-31 source is identified only after its restoration: correct, but not timely. The active-mass trace, the $\Delta M$ decomposition and the development counts are in the supplement.

### D. A CONTINUOUS SINGLE-STATE PHYSICAL CONTROL

A development control keeps one committed augmented physical state through restoration instead of competing histories. It uses the same nominal plant and measurement operators, profiles severity over short alternatives of at most 19 reports and applies the same six-report publication clock. On 54 development replays (three parents, three R/X conditions), this single-state generalized likelihood ratio (GLR) detector attains 37/54 timely decisions with 37 extras, against 44/54 and 6 for the nominal history bank and 51/54 and 3 for the five-model bank. The development schedules pack six interventions into 80-s records instead of 120 s, which likely explains why the nominal bank scores lower here than in Table 5. The three procedures also differ in onset search and severity scoring, so this comparison indicates, but does not isolate, the benefit of explicit recovery histories over a single

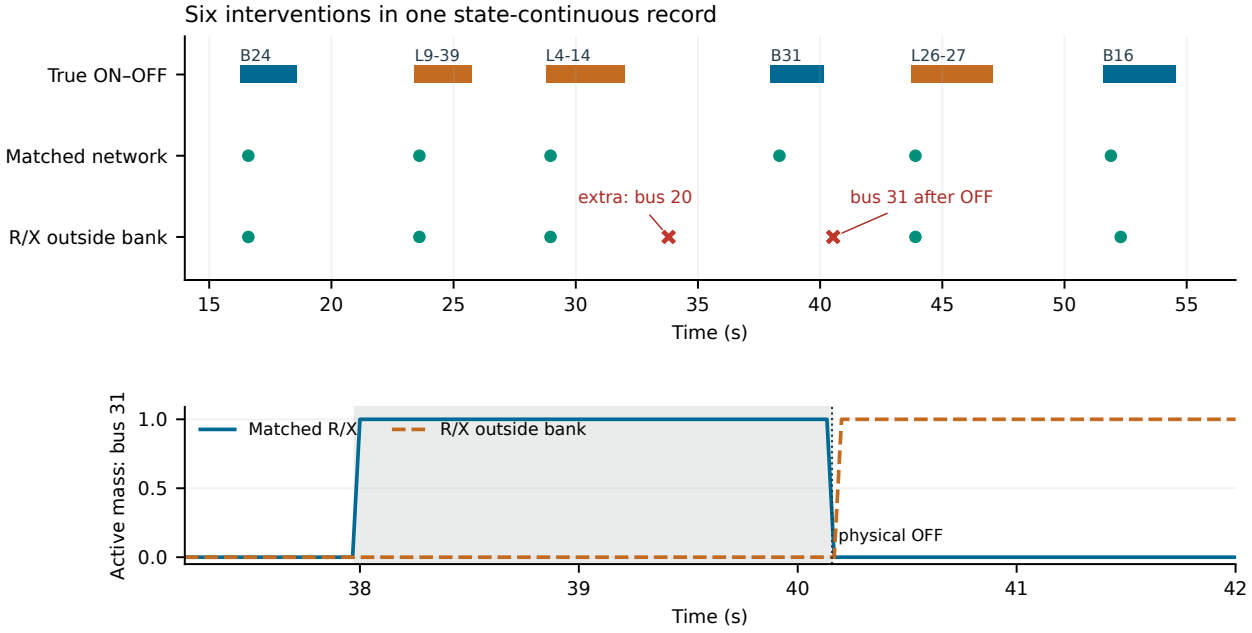



**FIGURE 9.** **One continuous development trajectory, shown for interpretation. Top: physical ON–OFF intervals and emitted diagnoses under matched and out-of-bank R/X. Bottom: bus-31 active mass with the true intervention shaded. A diagnosis after OFF misses the timely endpoint even when it identifies the source. True timing enters only this retrospective display.**

continuous state.

### *E. CONTROLLED SEVERITY-SCORE AND STATE-CARRY STUDY*

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank, we fixed a subsequent $2 \times 2$ study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{25}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{26}$$

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k / v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

#### 1) Component Outcomes

Table 7 and Fig. 10 report coverage and alarm cost for the four arms. With the state carried, marginal and bounded-profile severity scores tie on every reported endpoint. Resetting the 114 estimated physical-state deviations at inferred closure reduces timely attribution from 108/108 to 75/108 and raises the unmatched alarms from 1 to 166 under both scores. The reset leaves instrumental state, covariance regime, prior weights, thresholds and publication rules intact, and the simulated plant never resets.

Fig. 11 compares this outcome with Prediction 3. The spurious evidence exceeds $\lambda^\star$ after every intervention, by at least two and up to seven orders of magnitude, and the reset arm is harmed after 84 of the 108 interventions: an extra alarm

**TABLE 7. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.**

| Procedure | Timely | Extras | Normal FA | False restorations | No timely | Delay (s) |
|---|---|---|---|---|---|---|
| Marginal + carry | 108/108 | 1 | 0/1,764 s | 1 | 0/108 | 0.267 |
| Profile + carry | 108/108 | 1 | 0/1,764 s | 1 | 0/108 | 0.267 |
| Marginal + reset | 75/108 | 166 | 0/1,764 s | 130 | 33/108 | 0.337 |
| Profile + reset | 75/108 | 166 | 0/1,764 s | 130 | 33/108 | 0.337 |

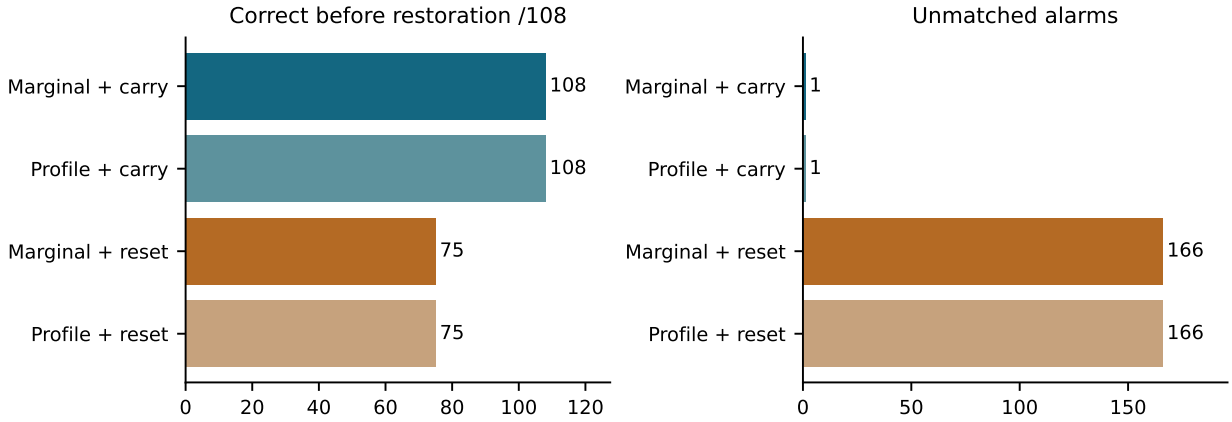



**FIGURE 10. Severity-scoring and state-carry study on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the alarm rate may differ between arms.**

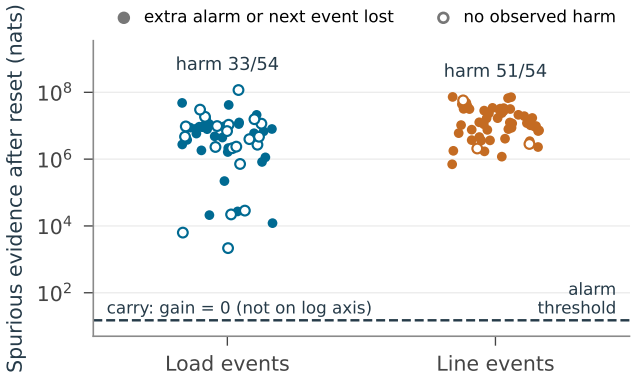



**FIGURE 11. Predicted spurious evidence after a state reset, (24), for each of the 108 interventions of the controlled study, against the observed outcome of the reset arm. Filled: an extra alarm in the following gap or a lost timely attribution of the next intervention; open: neither. With carry the predicted spurious evidence is zero.**

in the following gap or a lost timely attribution of the next one. Harm is almost certain after line events (51/54), whose openings excite larger electromechanical swings, and occurs after 33/54 load events. The bus-31 events carry the smallest spurious evidence and are harmed after three of seven. The analysis therefore explains why forgetting the state fails at the population level, while the outcome of an individual load event, given evidence far above threshold, is decided by the episode and publication rules rather than by the size of the evidence. These are within-simulator contrasts at the original thresholds; the two scores are compared at equal thresholds, not at equal alarm rates.

### *F. NOISE STRESS WITH UNCHANGED CODE*

To test Predictions 1 and 2 outside the calibrated noise, we reran the nominal bank without any change to its equations, thresholds or rules on four held-out calibration event records (12 load and 12 line interventions, matched network) and two 600-s normal records (0.33 h). Six noise scenarios were generated with the simulator's own noise stream: the original realization scaled by $\times 1$, $\times 5$, $\times 10$ and $\times 20$, and the original plus independent white per-PMU phasor noise with TVE-equivalent standard deviation 0.05% and 0.1%, propagated to frequency and ROCOF by the simulator's 21-report estimator. The $\times 1$ scenario reproduces the original predictions message for message. The bank re-estimates its noise scale and observation discrepancy from the noisy 300-report prefix; the operating point is kept at its nominal-noise estimate because derivative matrices exist only there. The implementation refuses to start when the estimated noise scale leaves its calibrated support of $[0.1, 10]$; for this test only, that bound was widened, and every other line of code is unchanged.

Table 8 and Fig. 12 give the outcomes. Scaling the calibrated noise up to twenty times leaves line attribution intact and loses one load intervention, a 4.1% change at bus 26, at $\times 5$ and $\times 10$ but not at $\times 20$. Prediction 2 classifies that change as detectable at all three levels, so the miss reflects the episode rules rather than missing information. The median load alarm delay moves only from 0.35 to 0.36 s, and no scenario produces an extra alarm on event records or any alarm on the normal records.

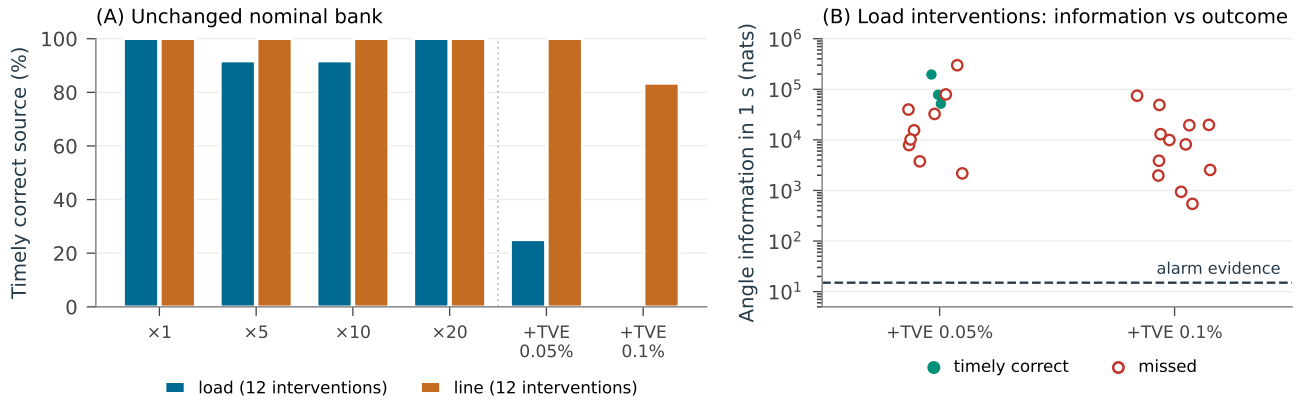



**FIGURE 12.** **Nominal bank under increased PMU noise, with code, thresholds and rules unchanged. (A) Timely correct family and source by scenario: the calibrated noise scaled by ×1–×20 (left of the dotted line) and the original noise plus independent white per-PMU phasor noise of 0.05% and 0.1% TVE (right). (B) For each load intervention under added independent noise, the information that a matched filter on cross-PMU-centered voltage and current angles accumulates in 1 s, against the observed outcome; the dashed line is the evidence required for an alarm.**

**TABLE 8.** **Noise-stress outcomes of the unchanged nominal bank: timely correct attributions, unmatched alarms on event records, and alarms on the two normal records.**

| Scenario | Load | Line | Extras | Normal |
|---|---|---|---|---|
| ×1 | 12/12 | 12/12 | 0 | 0 |
| ×5 | 11/12 | 12/12 | 0 | 0 |
| ×10 | 11/12 | 12/12 | 0 | 0 |
| ×20 | 12/12 | 12/12 | 0 | 0 |
| +TVE 0.05% | 3/12 | 12/12 | 0 | 0 |
| +TVE 0.1% | 0/12 | 10/12 | 0 | 0 |

Independent per-PMU noise behaves differently: the bank attributes 3/12 load interventions at 0.05% TVE and 0/12 at 0.1%, while lines remain at 12/12 and 10/12. These failures are not information-limited. For the same twelve load interventions, a matched filter on the cross-PMU-centered voltage and current angles, with the added independent noise alone, accumulates at least $2.2 \times 10^3$ nats in one second at 0.05% TVE and $5.4 \times 10^2$ at 0.1%, 36 times the alarm evidence or more. The implemented filter, however, adapts a single noise scale per signal family and keeps the cross-PMU correlation of the reference data, in which angle noise is almost entirely common to all units. Independent noise therefore raises its estimated angle-noise scale about 48-fold and suppresses the angle information on which load attribution relies; lines, whose signatures are two to three orders of magnitude larger (Fig. 5), survive. The test confirms Predictions 1 and 2 for the noise structure the bank models, and it identifies estimation of the cross-PMU noise covariance from the startup prefix as the extension required for field PMUs.

## VIII. DISCUSSION

### A. WHAT THE PHYSICS EXPLAINS

The information analysis turns the headline accuracy from an empirical observation into a consequence of the network equations. Under the noise of the reference data, every catalog intervention carries hundreds to tens of millions of times the evidence needed for an alarm within one second, so detection is limited by the publication rule rather than by information. Line alarms arrive at the confirmation floor, as predicted. Load alarms arrive about 4 reports later than the information allows. This latency is not explained by the information and is therefore algorithmic; the likely sources are the onset-proposal screen and the competition among severity intervals, which could be changed without new measurements. The errors that remain occur under out-of-bank R/X, where the predicted signature itself is wrong. The same analysis predicts where attribution should stop working, at the bus-31 load once the noise grows by a factor of about 7.4, and why forgetting the physical state is costly everywhere: the free recovery response projected on the PMUs is two to seven orders of magnitude larger than the alarm evidence. Physical-state continuity is therefore not a tuning choice but a requirement implied by the size of that response.

### B. RELATION TO LEARNED AND STATISTICAL DETECTORS

The learned adapters and the physical bank differ in how they obtain source responses: from labeled trajectories, or from the DAE operators. The adapters reproduce the line responses but miss most load changes, a pattern already present in their development validation. The comparison therefore shows that a physically derived dictionary attributes small load changes that these adapters do not, at the cost of more alarms when the network model is wrong. It does not establish superiority over a learned detector calibrated to a common alarm budget. Similarly, the adapter of Chen *et al.* fails here because its quasi-static white-noise model does not match the plant, not because quickest-change detection is unsuitable; an information-matched version of that test would use the switched-DAE signatures of Section V.

### C. LIMITS OF THE EVIDENCE

*Shared model family.* Observations and predictors come from the same research plant. Nonlinear switching, noise and out-

of-bank R/X challenge the linearized inverse model, but transfer to a different dynamic model or to field data is untested. The nine load and nine line candidates were fixed during method development; they form a declared catalog, not a sample of all assets or event families.

*Measurement realism.* The noise model reproduces the reference data, whose differential angle noise is small. Independent per-PMU noise of 0.05–0.1% TVE, well inside the standard's limits, suppresses most load attributions of the implemented bank even though the angles still carry ample information (Section VII-F). Field use therefore requires estimating the cross-PMU noise covariance, not only its scale, from the startup prefix; this extension was not tested here. The terminal map retains a provisional BUS5 current terminal and excludes the BUS39 current angle; it is a research measurement hypothesis, not a verified installation. Cross-family noise covariance is not applied.

*Approximation.* Pruning, steady-state covariances, plug-in startup estimates and mean compression at closure make the bank approximate; its masses are attribution weights, not calibrated probabilities. Retrospective proposals update the history posterior but not the network-model weights. The birth and restoration hazards are conservative relative to the test schedules; their effect is bounded by (22) but was not swept.

*Operation.* The startup prefix must be event-free, all PMUs available and the noise within the implemented support. The nominal bank runs faster than the input on one thread; the five-model bank does not. These results support diagnostic development, not protection or control deployment.

## IX. CONCLUSION

A restored electrical intervention can remain the cause of the observed response. We formulated load/line attribution from sparse PMUs as change detection and isolation with switched-DAE post-change models, in which histories start from estimated past states, keep that state through restoration and separate the active intervention from its originating cause. Because the histories are linear-Gaussian given their mode paths, the network operators also give each candidate's detection and isolation information, its minimum detectable change and the cost of forgetting the physical state.

In an independent simulated evaluation the bank attributes 644/648 interventions before restoration and raises no alarm in 11.8 h of normal records; its remaining errors and most of its 72 extra alarms arise from network-model mismatch. The information analysis explains this level of performance, separates the information-limited part of the alarm delay from its algorithmic part, and predicts the cost of resetting the state and the noise level at which the smallest load becomes undetectable. With unchanged code the bank tolerates a twentyfold increase of the calibrated noise. Independent per-PMU noise defeats its scalar noise adaptation while the angle signatures still carry ample information, so the cross-PMU noise covariance, not the physics, is the next element to estimate. The single-model bank already runs faster than the PMU reporting rate.

Two directions follow. The detectability analysis can be extended from a fixed catalog to placement and catalog design, since it ranks sources by information before any data are collected. The same framework accommodates other parameter-override events, such as generation changes, and measurement-integrity hypotheses that are inconsistent with network dynamics; testing them, together with transfer to an independent dynamic model, is the next step toward field use.

## REFERENCES


[1] I. V. Nikiforov, "A generalized change detection problem," *IEEE Transactions on Information Theory*, vol. 41, no. 1, pp. 171–187, 1995.

[2] T. L. Lai, "Sequential multiple hypothesis testing and efficient fault detection-isolation in stochastic systems," *IEEE Transactions on Information Theory*, vol. 46, no. 2, pp. 595–608, 2000.

[3] A. Tartakovsky, I. Nikiforov, and M. Basseville, *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. Boca Raton, FL: CRC Press, 2014.

[4] A. S. Willsky and H. L. Jones, "A generalized likelihood ratio approach to the detection and estimation of jumps in linear systems," *IEEE Transactions on Automatic Control*, vol. 21, no. 1, pp. 108–112, 1976.

[5] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.

[6] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.

[7] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.

[8] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023.

[9] A. N. Shiryaev, "On optimum methods in quickest detection problems," *Theory of Probability and Its Applications*, vol. 8, no. 1, pp. 22–46, 1963.

[10] B. Friedland, "Treatment of bias in recursive filtering," *IEEE Transactions on Automatic Control*, vol. 14, no. 4, pp. 359–367, 1969.

[11] H. A. P. Blom and Y. Bar-Shalom, "The interacting multiple model algorithm for systems with markovian switching coefficients," *IEEE Transactions on Automatic Control*, vol. 33, no. 8, pp. 780–783, 1988.

[12] Y. Boers and H. Driessen, "A multiple model multiple hypothesis filter for Markovian switching systems," *Automatica*, vol. 41, no. 4, pp. 709–716, 2005.

[13] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.

[14] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.

[15] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.

[16] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.

[17] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.

[18] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.

[19] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.

[20] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE*

*Journal of Selected Topics in Signal Processing*, vol. 8, no. 6, pp. 1140–1153, 2014.
[21] R. Anguluri, O. Kosut, and L. Sankar, "Localization and estimation of unknown forced inputs: A group LASSO approach," *IEEE Transactions on Control of Network Systems*, vol. 10, no. 4, pp. 1997–2009, 2023.
[22] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.
[23] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.
[24] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.
[25] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.
[26] R. Nagi, X. Huan, and Y. C. Chen, "Bayesian inference of parameters in power system dynamic models using trajectory sensitivities," *IEEE Transactions on Power Systems*, vol. 37, no. 2, pp. 1253–1263, 2022.
[27] A. Alalem Albustami, A. F. Taha, and S. Mahadevan, "DAE-aware bayesian inference for joint generator-network parameter estimation," In press, Electric Power Systems Research; arXiv:2604.15686, 2026. [Online]. Available: https://arxiv.org/abs/2604.15686
[28] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.
[29] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
[30] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.
[31] A. Pomarico, A. Berizzi, and J. N. Kutz, "A shallow recurrent decoder for dynamic state estimation with a limited number of PMUs in power systems," *arXiv preprint arXiv:2607.00116*, 2026.
[32] A. Dey, S. Chakraborty, and M. V. Salapaka, "Topology change estimation in power distribution networks: A machine learning approach," *IEEE Transactions on Smart Grid*, vol. 17, no. 4, 2026.
[33] IEEE, "IEEE standard for synchrophasor measurements for power systems," IEEE Std C37.118.1-2011, 2011.



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*Digital Object Identifier*

# Supplementary Material for Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

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**ABSTRACT** This supplement documents the implemented recovery-aware history bank: its local filtering calculations, measurement map, publication rules and simulation contracts. It records the complete confirmatory outcomes, controlled severity-score/state-carry ablation and continuous single-state control. The operating-point signature and recovery-geometry studies keep their separate, known-onset contracts and denominators. All empirical results reuse the fixed evidence; no new campaign or threshold selection is introduced here. Earlier voltage-reconstruction, simultaneous-source and general diagnosability studies are preserved in the separately compiled Historical Technical Notes.

**INDEX TERMS** phasor measurement units, sequential diagnosis, physical state continuity, conditional filtering, reproducibility.

## APPENDIX A IMPLEMENTED PREDICTOR AND CONDITIONAL UNCERTAINTY

This section specifies the local operators and conditional distributions used by the fixed continuous estimator. It distinguishes identities of that local model from approximations introduced by its finite history bank. The numerical checks below exercise these identities on controlled inputs; they do not recalibrate or rerun the reported evaluation.

### A. AFFINE OPERATORS AT A CONSISTENT TOPOLOGICAL ANCHOR

For network hypothesis $\ell$, let $(\bar{x}_\ell, \bar{u}_\ell)$ be the operating point determined by the prefix-fitted load and dispatch factors. Fix its differential coordinates while solving the algebraic constraints separately for each topology $q$:

$$g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0. \tag{1}$$

The resulting point need not be a differential equilibrium after a line opening. Assume a nonsingular algebraic Jacobian and a fixed angle gauge. In the expressions below, $f$ is the differential vector field after division by any nonunit differential mass entries. All derivatives are evaluated at the topology-specific point in (1); the network index is suppressed for readability. Define

$$\begin{aligned} c_q &= f_q(\bar{x}, \bar{z}_q, \bar{u}), \\ A_q^c &= f_x - f_z g_z^{-1} g_x, & B_q^c &= f_u - f_z g_z^{-1} g_u, \\ C_q^c &= H_x - H_z g_z^{-1} g_x, & D_q^c &= H_u - H_z g_z^{-1} g_u, \\ d_q^{el} &= H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u}). \end{aligned} \tag{2}$$

For the admitted scalar load direction $\delta u = La$, the continuous approximation is $\dot{\delta x} = A_q^c \delta x + c_q + B_q^c La$ and $\delta y = C_q^c \delta x + d_q^{el} + D_q^c La$. For a line, the topology selects the operator and the amplitude direction is zero. A nonzero pre-event state contributes through $\delta x$; $d_q^{el}$ supplies the zero-deviation offset at the fitted anchor. This is a local expansion, so large excursions can still cause prediction error even when the algebraic solve and derivatives are accurate.

For a mode lasting $h$, compute the affine flow without inverting $A_q^c$:

$$\mathcal{A}_q = \begin{pmatrix} A_q^c & c_q & B_q^c L \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$
$$\exp(h\mathcal{A}_q) = \begin{pmatrix} F_q(h) & \beta_q(h) & U_q(h) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3}$$

Thus $\beta_q(h) = \int_0^h \exp(A_q^c s) c_q \, ds$. If an admitted switch divides one report into an old-mode interval $h_1$ and a new-mode interval $h_2$, then

$$F = F_2 F_1, \qquad \beta = F_2 \beta_1 + \beta_2, \qquad U = F_2 U_1 + U_2. \quad (4)$$

The physical state is continuous across this composition. The final-mode measurement map uses the consistent algebraic right limit. The predictor then appends phase history and the deterministic memory of the instrumental transformation. Scaling these coordinates and applying the same affine augmentation to (4) produces the stored $T_o, J_o, g_o, u_o, j_o, v_o$ operators. In their report ordering,

$$\begin{aligned} r_k &= J_o \xi_k + j_o + v_o a + \epsilon_k, \\ \xi_{k+1} &= T_o \xi_k + g_o + u_o a + \eta_k. \end{aligned} \quad (5)$$

The measurement is scored before correcting the state for the next report. The constant startup discrepancy is applied to the observations; it does not replace the topology-dependent affine terms.

The construction has three steps. A static solve of (1) exports the network forcing, the PMU reference and their reduced Jacobians; the conditional builder forms the topology offsets, the mode flows and the one-report operators; and a final check verifies that the constructed reference matches the prefix-fitted operating point. The reproduction manifest preserves these files and their hashes. No event truth is used to select the startup operating point.

### B. FOUR AMPLITUDE INTERVALS REPRESENT ONE PRIOR

Let $I = [-0.2, 0.2]$ be partitioned into four disjoint intervals $I_j$ up to endpoints of zero probability. Write $Z_J(\mu, v) = \int_J \mathcal{N}(a; \mu, v)\, da$ for any interval $J$. The initial prior has the exact decomposition

$$\begin{aligned} p_0(a) &= \sum_{j=1}^{4} \alpha_{j,0} \, \mathrm{TN}_{I_j}(a; \mu_0, v_0), \\ \alpha_{j,0} &= Z_{I_j}(\mu_0, v_0) / Z_I(\mu_0, v_0). \end{aligned} \quad (6)$$

The implementation assigns one history per source, proposed onset, restoration path, and interval. Each retains its own $(m_{h,j}, d_{h,j}, \mu_{h,j}, v_{h,j}, w_{h,j})$; there is no per-report collapse to a single amplitude distribution. The four interval histories use the same physical operators. They are neither quadrature nodes nor four separately linearized severity models.

For a fixed physical history with observations $Y$, let $Z_j(Y) = \int_{I_j} p(Y \mid a, h)\, \mathrm{TN}_{I_j}(a; \mu_0, v_0)\, da$. Then

$$p(Y \mid h) = \sum_j \alpha_{j,0} Z_j(Y), \qquad \alpha_j(Y) = \frac{\alpha_{j,0} Z_j(Y)}{p(Y \mid h)}. \quad (7)$$

Applying the local predictive integral sequentially gives the same identity. Identical linear operators and gains preserve the common untruncated Gaussian parameters for interval copies of the same path; their truncated moments and normalized weights differ. Subsequent beam pruning can remove interval histories, so equivalence with an unpartitioned bank is not guaranteed after pruning. The recorded discarded mass describes that approximation and is not a bound on all future decisions.

### C. AMPLITUDE DEPENDENCE AND MIXTURE COMPRESSION

Condition on a retained history and the reports used so far. For $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, set $\sigma = \sqrt{v}$, $\alpha = (l-\mu)/\sigma$, $\beta = (u-\mu)/\sigma$, $Z = \Phi(\beta) - \Phi(\alpha)$, and $\lambda = [\phi(\alpha) - \phi(\beta)]/Z$. Its moments are

$$\begin{aligned} \bar{a} &= \mu + \sigma\lambda, \\ V_a &= v \left[ 1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{Z} - \lambda^2 \right]. \end{aligned} \quad (8)$$

The truncated-moment routine evaluates the normalizer in the log domain and bounds small numerical excursions of the moments. The Gaussian parameters $\mu, v$ are therefore distinct from the conditional amplitude mean and variance.

For the affine conditional state model $\xi \mid a, h \sim \mathcal{N}(m_h + d_h a, P_h)$, total expectation and covariance give

$$\begin{aligned} \bar{\xi}_h &= m_h + d_h \bar{a}_h, \\ \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ V_h &= P_h + d_h V_{a,h} d_h^\top. \end{aligned} \quad (9)$$

With normalized history weights $\omega_h$, including the no-new-event history, the complete mixture moments would be

$$\begin{aligned} \bar{\xi} &= \sum_h \omega_h \bar{\xi}_h, \\ P_{\mathrm{mix}} &= \sum_h \omega_h \left[ V_h + (\bar{\xi}_h - \bar{\xi})(\bar{\xi}_h - \bar{\xi})^\top \right]. \end{aligned} \quad (10)$$

This identity applies within a network hypothesis and its common state coordinates. It does not justify averaging absolute rotor states between different network equilibria.

The continuous estimator uses the shared regime covariance $P_\rho$ in place of $P_h$. It updates the affine means and amplitude distributions, and computes marginal diagonal state variances for diagnostics. At closure it carries $\bar{\xi}$, clears explicit histories, and records the trace of amplitude/mean spread omitted from the new single-history representation. It does not carry the complete covariance in (10), nor the uncertainty of the fitted startup context. Instrumental and discrepancy states remain in the carried augmented mean.

For comparison, an exact conditional covariance recursion for the local linear model (5), with independent noises of covariances $R, Q$, is

$$\begin{aligned} S_h &= J_o P_h J_o^\top + R, \\ K_h &= T_o P_h J_o^\top S_h^{-1}, \\ P_h^+ &= T_o P_h T_o^\top + Q - K_h S_h K_h^\top. \end{aligned} \quad (11)$$

This is a diagnostic reference, not a description of a new executed continuous estimator. The fixed implementation solves the Riccati equation for the nominal operator in each

**TABLE 1. Deterministic local-model checks. Maximum absolute errors over 72 scalar updates, five switch positions, and 12 joint Gaussian updates; each row uses its stated reference.**

| Check | Maximum error |
|---|---|
| Log evidence: partition | $1.78 \times 10^{-15}$ |
| Amplitude mean/variance | $4.16 \times 10^{-17}$ |
| Log evidence: quadrature | $3.11 \times 10^{-15}$ |
| Affine flow: ODE reference | $2.22 \times 10^{-16}$ |
| Joint mean/covariance | $5.55 \times 10^{-17}$ |

discrepancy regime and uses that shared covariance for the admitted switched operators.

Sequential likelihood factorization conditions on past reports; it does not require marginal independence of the observations. The instrumental VAR/AR filter and the latent discrepancy AR state already model temporal dependence. Remaining covariance misspecification, linearization error, adaptive proposals, and compression can nevertheless affect the resulting weights. Their empirical calibration requires pre-update residual and coverage diagnostics, separately from the algebraic checks below.

### D. DETERMINISTIC VERIFICATION AND ITS BOUNDARY

A deterministic check loads the implemented amplitude routine without running a campaign. It compares sequential one-interval and four-interval evidence and moments with direct scalar quadrature, including zero information and amplitudes on interval boundaries. An independent small switched linear system checks affine-flow composition against numerical integration. A linear Gaussian reference also compares the conditional affine-state recursion with a joint state–amplitude Kalman recursion for an untruncated Gaussian amplitude; the truncated case is checked separately by quadrature. All histories are retained in these checks. Source hashes, tolerances, and complete results accompany the generated table. These checks establish local-model identities on the stated inputs. They establish neither calibration of the pruned network bank nor robustness of its nonlinear physical approximation, and add no event outcomes to the evaluation cohorts.

## APPENDIX B HOW TO READ AND REPRODUCE THE CURRENT METHOD

This companion begins with the recovery-aware causal method used in the current article. Sections B–D give its execution order, local derivations, worked examples and simulation contracts. The separately compiled Historical Technical Notes preserve full diagnosability proofs, earlier state-reconstruction and batch-estimator studies, and secondary evidence. The earlier studies support particular components; they are not additional tests of the complete causal algorithm.

The article retains the physical model, the defining evidence integral, the recovery mechanism, the decision outputs and decisive results. This supplement supplies the intermediate calculations. The Historical Technical Notes develop functional observability, nuisance-profiled information and source ambiguity, and distinguish the implemented batch estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint estimator $\mathcal{E}_{\mathrm{J}}$. Neither should be silently substituted for the causal physical history bank. The conditional Gaussian updates use standard filtering identities [1]; the derivation below specifies their role and the approximations made by the implemented bank.

### A. INPUTS, OUTPUTS AND THE INFORMATION BOUNDARY

The inputs are successive reports from eight PMUs, the fixed physical and instrumental operators, the admitted load/line catalog and the fixed method configuration. The output is a timestamped sequence of detection, family/source and restoration messages, together with retained attribution masses. True ON/OFF times, true sources and generating severities belong to the evaluator. They are not inputs to the causal algorithm.

Three times must remain distinct: the latest observed report, the estimated physical onset, and the public emission time. Looking back to propose an onset is compatible with causality when the message is emitted only after the required reports arrive. It does not permit backdating the public alarm.

### B. EXECUTION ORDER FOR ONE STREAM

1) **Load the fixed operators.** Retain voltage reference, oriented terminal currents, load input directions, switch modes and instrument memory. A line opening changes the network and observation operators. It is not represented by fitting a load amplitude.
2) **Fit the startup context.** Use the first 300 reports for operating factors, common reference and noise scale. For each network hypothesis, set the voltage/current discrepancy to the difference between the observed prefix mean and its reference prediction. Keep that offset fixed afterwards. Frequency and ROCOF use their empirical prefix centers. The prefix is not scored again as independent model-selection evidence.
3) **Predict the next report.** Propagate the retained histories and evaluate each network model's predictive density before correcting states or choosing new histories with the current report.
4) **Propose candidate onsets.** The fixed screen uses lags 0–18 reports within a ten-second stored state history. Each proposal starts from a stored state before its proposed onset, including any recovery already present. The bank does not enumerate every possible onset in the stored interval.
5) **Score histories.** For loads, integrate severity with the bounded prior and the local conditional Gaussian likelihood. For lines, score the admitted switched operator. Score a report before using that same report for its conditional state correction.
6) **Update and compress.** Update severity and conditional states, network weights, restoration children and the retained beam. Record discarded mass. At episode closure retain the augmented-state mixture mean; the

omitted between-history spread is a documented approximation, not an exact posterior covariance.

7) **Publish and record.** Aggregate active mass $M$, cause mass $\Gamma$ and $\Delta M$. Apply the fixed detection, diagnosis and restoration rules. Save emission time and the last report used, in addition to estimated onset and source. Continue propagation through OFF.

### C. WHY REPLAYED REPORTS ARE NOT NEW EVIDENCE

Let a retained parent have log score $\ell_{0:k} = \ell_{0:\tau-1} + \ell^{old}_{\tau:k}$. Replacing its suffix by a newly proposed physical history gives

$$\ell^{new}_{0:k} = \ell_{0:k} - \ell^{old}_{\tau:k} + \ell^{new}_{\tau:k}. \tag{12}$$

Transition and prior terms must accompany their corresponding history. The rolling screen supplies a proposal, not an extra independent likelihood. Adding the new suffix without removing the old suffix would count the same reports twice. This bookkeeping statement does not make adaptive screening or beam pruning exact Bayesian inference.

### D. FIXED NUMERICAL CHOICES

The input grid is 30 reports/s. The five network hypotheses are nominal, global $\pm 2\%$ and selected-branch $\pm 3\%$ paired R/X perturbations. Each hypothesis is fixed in time; the weights change. They do not identify a continuous impedance posterior.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$. Birth mass is divided equally between load and line families, then uniformly among the admitted sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over intervals bounded by $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can create OFF children at either half-report position, inheriting state and severity. After likelihood normalization, the beam retains the union of the 96 highest-weight recent histories, 64 older histories and at least one per occupied source. The union is renormalized.

The internal observer confirms active-mass evidence at 0.99 for six reports and has a separate 15-report internal OFF clock. The public physical-bank publisher requires an internal pending episode and six reports of source-qualified active support above 0.99; its public OFF requires six reports of the diagnosed cell's joint cause-and-restoration mass above 0.99. The detailed source qualification, diagnosis and replacement rules are given below. These thresholds are implementation settings, not guarantees of probability calibration or false-alarm control. Nominal and mixture use their fixed publication paths, so their comparison does not isolate network averaging alone. The admission-rule ablation changes publication, not the recovery dynamics.

### E. SOURCE QUALIFICATION AND PUBLIC EPISODE RULES

The source screen and the onset search are distinct. Source qualification uses a 30-report (one-second) conditional innovation vector $v$ with 47 coordinates per report. Fixed normal-mode gains propagate the stored pre-window state and whiten the suffix. For each prepared response $s_j$ and lag from 0 to 25 reports, let

$$\begin{aligned} g_j &= s_j^\top s_j, \qquad u_j = s_j^\top v, \qquad R = v^\top v, \\ \widehat{a}_j &= \mathrm{clip}(u_j / \max(g_j, 10^{-30}), -0.2, 0.2), \\ G_j &= 2\widehat{a}_j u_j - \widehat{a}_j^2 g_j, \\ F_j &= \frac{\max(G_j, 0)}{\max\{(R - G_j)/1409, 1\}}. \end{aligned} \tag{13}$$

Line columns use fixed amplitude one. The source score is the maximum over its columns; the threshold is the $1 - 0.001/936$ quantile of an $F_{1,1409}$ distribution, approximately 24.0115. This threshold and the variance floor are a fixed proposal/publication rule, not a calibrated global false-alarm probability. The statistic is never multiplied into the Bayesian history likelihood. This one-second qualification screen does not enlarge the separate birth search beyond its 0–18 report lags.

Define $b_c = 1$ for a line, and $b_c = \mathbf{1}\{F_c > 24.0115\}$ for a load. At each monitored report, the publisher updates consecutive-report counters for $M_c b_c \geq 0.99$, $\Gamma_c \geq 0.99$, and $\Gamma_c P(\mathrm{OFF} \mid c) \geq 0.99$. A false condition clears its counter. An internal alarm opens a pending episode. A public alarm then requires either six pending reports with $\sum_c M_c b_c \geq 0.99$ or an already accumulated six-report line-active counter. The latter permits line evidence collected before the internal pending message to count.

A public episode receives one diagnosis when its highest cause-mass cell has six consecutive reports above 0.99, or at the 15-report deadline after the public alarm. The output family maximizes summed cause mass; sources are ranked within that family. The diagnosed source becomes the episode owner. Six reports of that owner's joint cause-and-OFF mass above 0.99 emit a public restoration. These publication operations do not rescale $M$ or reset the physical state.

The internal mixture lifecycle separately replaces an episode after six reports with mass at least 0.99 on active histories whose onset is later than its alarm (using the implementation's quarter-report timing guard). It withdraws an unconfirmed episode after the diagnosis delay when null mass has exceeded 0.99 for 15 reports. Its internal restoration requires cause mass at least 0.99 and conditional retired mass at least 0.99 for 15 reports. These terminal messages are propagated to the public episode if already emitted; an unconfirmed pending episode is recorded as rejected. Finalization marks any remaining public episode right-censored. The nominal comparator retains its own fixed internal lifecycle, so the nominal/mixture comparison is not an isolated model-averaging ablation.

## APPENDIX C LOCAL CALCULATIONS AND WORKED EXAMPLES

### A. COMPLETING THE BOUNDED-AMPLITUDE INTEGRAL

For one retained severity interval, write $e = sa + \epsilon$, with $\epsilon \sim \mathcal{N}(0, S)$, $S \succ 0$, and $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, $v > 0$. Here $\mu, v$ are parameters of the underlying untruncated Gaussian, not the bounded distribution's moments. With $g = s^\top S^{-1} s$ and $q = s^\top S^{-1} e$, expansion gives

$$\begin{aligned}(e - sa)^\top S^{-1}(e - sa) + (a - \mu)^2/v \\ = e^\top S^{-1} e + \mu^2/v - \mu'^2/v' + (a - \mu')^2/v', \qquad (14)\\ v' = (v^{-1} + g)^{-1}, \qquad \mu' = v'(\mu/v + q). \qquad (15)\end{aligned}$$

Define $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Integrating the last Gaussian square over $[l, u]$ yields

$$\begin{aligned}\log L = & -\tfrac{1}{2}[e^\top S^{-1} e + \log|S| + n_y \log(2\pi)] \\ & + \tfrac{1}{2}[\log(v'/v) + \mu'^2/v' - \mu^2/v] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \qquad (16)\end{aligned}$$

The posterior on that interval is $\mathrm{TN}_{[l,u]}(\mu', v')$. With $\alpha = (l - \mu')/\sqrt{v'}$, $\beta = (u - \mu')/\sqrt{v'}$, $Z' = \Phi(\beta) - \Phi(\alpha)$ and standard normal density $\phi$, its mean is

$$\mathbb{E}[a \mid e] = \mu' + \sqrt{v'}\,\frac{\phi(\alpha) - \phi(\beta)}{Z'}. \qquad (17)$$

Thus neither the untruncated mean $\mu'$ nor a best-fit severity can simply replace the integral. The implementation combines retained intervals using their weights. The identity is exact for this conditional Gaussian model; it does not remove linearization, shared-covariance or pruning error.

### B. NUMERICAL EXAMPLE OF ONE INTERVAL

This dimensionless example illustrates the calculation; it is not an IEEE-39 simulation or a new performance result. Set $S = 1$, $s = 2$, $e = 1$, $\mu = 0$, $v = 1$ and $[l, u] = [0, 1]$. Then $g = 4$, $q = 2$, $v' = 0.2$ and $\mu' = 0.4$. Substitution in (16) gives $L \simeq 0.342688$ and $\mathbb{E}[a \mid e] \simeq 0.464941$. The different posterior mean follows from truncation. A separate numerical check compares the closed form against direct numerical integration, including the posterior mean. It checks algebra only and does not modify a fixed experiment.

### C. CONDITIONAL STATE CORRECTION

Represent the conditional state mean as $m_h + d_h a$. Let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the stored one-report transition and observation operators for switch $o$. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With shared covariance $P_\rho$ in discrepancy regime $\rho$,

$$\begin{aligned}S_o = R + J_o P_\rho J_o^\top, \qquad K_o = T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ = T_o m_h + g_o + K_o e, \quad d_h^+ = T_o d_h + u_o - K_o s. \qquad (18)\end{aligned}$$

To see the last line, apply the correction to a fixed amplitude:

$$T_o(m_h + d_h a) + g_o + u_o a + K_o(e - sa) = m_h^+ + d_h^+ a.$$

The severity distribution is updated by the same likelihood integral. Regime covariances come from nominal-operator Riccati equations. The physical history bank does not propagate an independent exact covariance for every switched history. At closure the mean is retained and the omitted mixture spread recorded.

### D. RESTORATION IS NOT A STATE RESET

Consider the illustrative scalar system $\dot{x} = -x + u$, $y = x$, with $x(0) = 0$. Apply $u = a$ for $0 \le t < \kappa$ and restore $u = 0$ afterwards. Direct integration gives

$$x(t) = \begin{cases} a(1 - e^{-t}), & 0 \le t < \kappa, \\ a(1 - e^{-\kappa})e^{-(t-\kappa)}, & t \ge \kappa. \end{cases} \qquad (19)$$

Both expressions agree at $\kappa$. For $a = 1$ and $\kappa = 2$, $x(2) = 0.864665$ and $x(3) = 0.318092$ even though the intervention is OFF. Resetting the reference to equilibrium would discard that predictable recovery. This example explains state continuity; it does not claim that one exponential describes the multimode IEEE-39 response.

### E. ACTIVE MASS, CAUSE MASS AND MODEL REWEIGHTING

For one model, suppose an illustrative cell has mass 0.60 in active histories and 0.30 in restored histories, with 0.10 elsewhere. Its active mass is $M_c = 0.60$, while its originating-cause mass is $\Gamma_c = 0.90$. Recovery can therefore remain attributable without declaring the input active. These assigned masses illustrate definitions, not calibrated probabilities from an experiment.

For two reports, adding and subtracting $\sum_\ell w_{\ell,k} M_{\ell,c,k-1}$ proves

$$\begin{aligned}\Delta M_c = & \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ & + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \qquad (20)\end{aligned}$$

For example, unchanged conditional masses $(0.8, 0.2)$ and network weights changing from $(0.5, 0.5)$ to $(0.75, 0.25)$ move $M_c$ from 0.50 to 0.65. The entire 0.15 change comes from model reweighting. A rising aggregate mass need not mean that each model found stronger within-model evidence.

## APPENDIX D SIMULATION SCENARIOS AND REPRODUCTION ORDER

### A. PLANT, MEASUREMENT AND PROVENANCE

The simulator is a Julia/PowerDynamics research model of the IEEE 39-bus system, not an officially validated equivalent of the competition reference record. Its contract fixes Julia 1.11.9, PowerDynamics 5.0.0, NetworkDynamics 1.3.0 and IEEE39 0.1.0. Initial states, effective parameter vectors, terminal map, reporting operator and stochastic model come from that snapshot. Nominal inventory files do not replace the effective operating point. The load law is the fixed impedance law.

The voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. Each current is an oriented branch-terminal measurement, not a

nodal injection. BUS5 retains the provisional historical 5–6 branch, terminal 5, hypothesis; producer CT identity is unconfirmed and BUS39 remains partially validated. Execution uses full-precision configuration files rather than rounded tables. The competition reference record provided retrospective calibration context only.

Each physical parent is integrated continuously through its interventions. Differential states remain continuous at switching, while algebraic values are recomputed consistently. Reports use $t = k/30$ with endpoints included; the switch-aligned report uses the post-switch algebraic value. Solver steps are recorded separately and observations are not interpolated across a switch. The causal instrumental history is preserved through OFF.

### B. CONTINUOUS DEVELOPMENT AND CONFIRMATION

The two development campaigns each contain three 80-s event parents with six interventions and one 80-s normal parent. Crossing the four parents with three R/X conditions gives 12 physical trajectories, 18 unique interventions and 54 event realizations per campaign. Both are development evidence in the current article. Repeated conditions are not independent event draws.

The confirmatory evaluation prespecifies 18 continuous parents, 27 persistent-line parents (nine lines, three repetitions), and 12 normal parents. The three R/X conditions produce $3(18+27+12) = 171$ intended physical trajectories; two instrumental-noise scales yield 342 observed records. The continuous parents contain $18 \times 6 = 108$ unique interventions. Their six condition/noise realizations are repeated observations of those parents. The 120-s and 600-s grids contain 3601 and 18001 reports, respectively. Completed results appear in the dedicated confirmation tables.

Continuous evaluation pairs an alarm temporally before inspecting its predicted label. The first unmatched post-ON alarm is eligible up to the next ON, with the inherited 30-s cap. Timely attribution requires correct family and asset emitted before physical OFF. Unmatched alarms, false restorations and open episodes are retained. Persistent-line evaluation instead scores the first stopping decision, including early stops, wrong sources, omissions and censoring. The detector of Chen *et al.* is evaluated only there; it has no load or restoration output. Confidence intervals use 10,000 paired parent-cluster resamples, not independent PMU samples.

### C. SEPARATE SIGNATURE AND RECOVERY CONTROLS

The context-signature study uses a single load at buses $\{3, 7, 16, 24\}$, a 10-s record, known ON at 3 s and OFF at 6 s, and one scalar operating-context direction. It contains 57 dictionary/check trajectories and 85 test trajectories (80 interventions and five controls), with three common noise seeds producing 255 observed test records. The context and severity are estimated from measurements, while onset and family are supplied. The generating source is used to assess the response approximation and severity error. Those conditional metrics do not measure unknown-onset causal attribution.

The recovery-geometry control uses 38 trajectories. Prior load histories at buses 7 and 24, followed by delays of 0.5 and 2 s, give four recovery contexts. Three source-pair comparisons within each context give 12 correlated comparisons. The prior input is active from 2 to 3 s; the subsequent response is examined over 1.5 s. The supplied history, onset, family and two-candidate universe define an oracle control, with one uncertain prior-severity direction. It does not evaluate the causal physical method, unknown onset, event cardinality or operational false alarms.

### D. REPRODUCTION WITHOUT CHANGING THE EXPERIMENT

1) Read each campaign's fixed plan and manifest. Select the intended population before interpreting any denominator. Keep current causal, signature, recovery-control and historical batch studies separate.
2) Resolve the plant state, parameter and measurement files from the versioned snapshot. Every new simulation is a new run, and a new hypothesis a new candidate; earlier preparation steps are never rerun inside a fixed baseline.
3) Preserve ON/OFF continuity, report endpoints, noise seeds and the boundary between observations and reserved event truth. Run inference before releasing truth to the evaluator.
4) Retain solver/inference failures in intended denominators. Aggregate at the declared parent level and preserve the original comparators and publication rules. Do not retune from confirmation results.
5) Rebuild the two documents from the versioned tables and figures. The default build does not regenerate scientific artifacts or import new campaign results; refreshing artifacts is a separate operation.

The reproducibility package keeps this walkthrough, the campaign evidence and the earlier mathematical development, compiled separately as Historical Technical Notes. A successful PDF build verifies document consistency; it is not a new scientific validation.

## APPENDIX E EXECUTED MEASUREMENT MAP

### A. MEASUREMENT CONTRACT

The executed channel map is taken from the versioned research-plant snapshot. The eight voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. The internal voltage phasor is in per unit with unit gain and is exported as phase-to-neutral RMS volts using 345 kV/$\sqrt{3}$; voltage angle uses one common affine phase reference plus the stored relative offsets. The reference interpretation is observational: physical common drift and instrument clock drift are not separately identified. Terminal currents are converted to outward branch convention from the simulator's into-terminal observable, then mapped with the row-specific polarity, gain and phase bias. Current bases use $100\,\mathrm{MVA}/(\sqrt{3}\,V_{\mathrm{terminal}})$; the common voltage base is 199.186 kV.

**TABLE 2. Eight PMU rows of the executed measurement map. V is a bus voltage in p.u.; I is an oriented branch-terminal phasor. Gains and phase biases are applied to current channels. BUS5 retains the historical 5–6, terminal-5 hypothesis and its producer CT identity is unverified. BUS39 current phase is partially validated.**

| PMU | V bus | Edge | Branch | Terminal | Side | Polarity | Gain | Bias (°) | I unit | Provenance |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | 2 | 2–3 | 3 | dst | 1 | 0.9581 | -1.173 | A | Terminal hypothesis; CT unverified |
| 5 | 5 | 9 | 5–6 | 5 | src | 1 | 1.0361 | 0.312 | A | BUS5 5–6 historical hypothesis; CT unverified |
| 6 | 6 | 11 | 6–7 | 6 | src | 1 | 1.0077 | 1.495 | A | Terminal hypothesis; CT unverified |
| 10 | 10 | 17 | 10–11 | 11 | dst | -1 | 1.0041 | 2.224 | A | Terminal hypothesis; CT unverified |
| 19 | 19 | 26 | 16–19 | 19 | dst | 1 | 1.0384 | 0.481 | A | Terminal hypothesis; CT unverified |
| 22 | 22 | 34 | 21–22 | 22 | dst | 1 | 1.0189 | 0.094 | A | Terminal hypothesis; CT unverified |
| 29 | 29 | 44 | 28–29 | 29 | dst | -1 | 1.0316 | -6.170 | A | Terminal hypothesis; CT unverified |
| 39 | 39 | 1 | 1–39 | 39 | dst | -1 | 0.9298 | 30.053 | A | BUS39 partial phase validation |

**TABLE 3. Fitted ExtraTrees comparators used in the held-out streams. Both were trained for this study and are distinct from the original competition models.**

| Comparator | Features | Windows | Trees |
|---|---|---|---|
| ExtraTrees | 846 | 1, 5, 10 s | 400 |
| ExtraTrees short | 1457 | 1, 5, 10 s; 3, 8, 15 reports | 400 |

Frequency is computed as $60 + \dot{\theta}/(360)$ from the reported voltage angle, and ROCOF as $\ddot{\theta}/360$ using a causal 21-report quadratic estimate. The generator fit is robust and magnitude weighted; the predictor uses a fixed unweighted quadratic fit over the same history. Four warmup outputs are unavailable; the confirmatory evaluation uses a 10-s startup context and begins monitoring after 12 s. The 47 retained coordinates exclude the BUS39 current-angle coordinate. Noise multipliers 1 and 2 scale the standard deviation under the fixed stream. The stochastic model stores a cross-signal covariance field, but the executed report operator applies noise by signal family without that cross-family factor. These choices describe a simulated sensor configuration, not a verified eight-PMU CT/PT installation.

## APPENDIX F LEARNED COMPARATOR LINEAGE AND DECISION CONTRACT

Both models were fitted on 32 physical parents from four earlier development campaigns; validation used 8 parents and 16 observed records from a fifth campaign. All of these were development data for the comparator construction. The fitted label set contains normal operation and all 18 catalog sources. The training target is the active source; after restoration, including physical recovery, its label is normal. This is not a source-excluded transfer test.

Each report selects the largest tree vote, and six consecutive reports of the same label drive the episode interface; there is no separate probability threshold. Tree votes are uncalibrated; the 300-report prefix and the start of monitoring at report 360 are fixed. Model hashes, training metadata and the observed class set are kept in the reproducibility package.

## APPENDIX G COMPLETE CONFIRMATORY OUTCOMES AND DIAGNOSTIC AUDIT

The fixed test contains 171 physical trajectories and 342 observed records. Every method has an accounted output for every intended record; the solver-rejection and inference-failure ledgers are empty. The continuous, persistent-line and normal physical-parent counts are 18, 27 and 12. Three network conditions and two noise levels repeat these parents. Normal controls are shared between evaluation lanes, not doubled independent exposure. Tables below are generated directly from the stored summary, records, event matches and paired intervals.

### A. TIMELY COVERAGE VERSUS CLASSIFICATION

The additional confusion view in Fig. 1 is a posthoc descriptive audit. It uses the original evaluator's temporal episode association and the first public diagnosis; it does not rematch events using source labels. A diagnosis emitted after physical OFF is counted as unavailable for this view. All diagnoses that are timely and matched in these four procedures have the correct family and exact source. The large load gap therefore reflects whether a diagnosis is available before OFF, not a measured advantage in conditional label accuracy among timely outputs. Unmatched alarms remain in the separate alarm tables. Normal records are not mixed into this event-level matrix.

### B. MAINTAINING A SUSTAINED EVENT AFTER THE FIRST STOP

The first-stop score alone hides subsequent fragmentation. Although both physical banks and both learned comparators correctly identify all 162 persistent openings first, the learned comparators later return to an inactive label and issue additional alarms. Table 11 describes those already recorded outputs. It does not change the fixed first-stop outcome or extend Chen's one-stop contract into a recurrent monitor. Retained open episodes are appropriate when an intervention continues beyond the record.

### C. MEASUREMENT INTEGRITY IS A SEPARATE EXECUTED COMPONENT

The availability interface correctly identifies all four missing-data intervals at the first received report that reveals the loss. This is a received-report clock, not zero transport or

**TABLE 4. Continuous outcomes by network and noise stratum. Each cell is timely family/source decisions / intended events; unmatched event-stream alarms. Replays share physical parents.**

| Network | Noise | Physical bank | Nominal bank | ExtraTrees | ET short |
|---|---|---|---|---|---|
| Matched | 1 | 108/108; 1 | 108/108; 1 | 56/108; 6 | 60/108; 5 |
| Matched | 2 | 108/108; 0 | 108/108; 0 | 54/108; 4 | 58/108; 0 |
| In-bank | 1 | 108/108; 2 | 108/108; 3 | 56/108; 5 | 60/108; 5 |
| In-bank | 2 | 108/108; 1 | 108/108; 2 | 54/108; 4 | 57/108; 0 |
| Out-of-bank | 1 | 106/108; 38 | 105/108; 47 | 56/108; 5 | 60/108; 6 |
| Out-of-bank | 2 | 106/108; 30 | 106/108; 39 | 54/108; 5 | 57/108; 0 |

**TABLE 5. Continuous monitoring: unavailable intended records, normal-record alarm incidence, normal alarm counts and available normal exposure, false restorations and open episodes. Counts combine six repeated strata descriptively; normal parents are not independent across strata. Missing outputs remain intended failures.**

| Method | Unavailable | Normal incidence | Normal alarms | Exposure (h) | False OFF | Open at end |
|---|---|---|---|---|---|---|
| Physical 5-model | 0 | 0/72 | 0 | 11.760 | 58 | 0 |
| Physical nominal | 0 | 0/72 | 0 | 11.760 | 68 | 0 |
| ExtraTrees | 0 | 0/72 | 0 | 11.760 | 29 | 0 |
| ET short | 0 | 0/72 | 0 | 11.760 | 16 | 0 |

**TABLE 6. Conditional median diagnosis/restoration delays (s), kept within each paired stratum. These medians summarize matched emitted diagnoses/restoration messages with finite delay and must be read alongside omissions, unavailable records and censoring; – denotes no finite estimate. Logical delays do not measure computational throughput.**

| R/X condition | Noise | Physical 5-model | Physical nominal | ExtraTrees | ET short |
|---|---|---|---|---|---|
| Matched | 1 | 0.27/0.18 | 0.27/0.18 | 0.23/0.18 | 0.19/0.18 |
| Matched | 2 | 0.27/0.18 | 0.27/0.18 | 0.30/0.18 | 0.20/0.18 |
| In-bank | 1 | 0.27/0.18 | 0.27/0.18 | 0.23/0.18 | 0.19/0.18 |
| In-bank | 2 | 0.27/0.18 | 0.27/0.18 | 0.30/0.18 | 0.20/0.18 |
| Out-of-bank | 1 | 0.33/0.18 | 0.33/0.18 | 0.23/0.18 | 0.19/0.18 |
| Out-of-bank | 2 | 0.26/0.18 | 0.33/0.18 | 0.30/0.18 | 0.20/0.18 |

**TABLE 7. Persistent-line comparison on the common observations. Each cell gives correct first source / intended openings; pre-change stops / right-censored openings; mean restricted correct delay (s). Wrong first sources remain in the denominator. Restricted delay retains the horizon penalty for failures and censoring defined by the fixed evaluator.**

| R/X condition | Noise | Physical 5-model | Physical nominal | Chen *et al.* |
|---|---|---|---|---|
| Matched | 1 | 27/27; 0/0; 0.18 | 27/27; 0/0; 0.18 | 12/27; 11/0; 54.58 |
| Matched | 2 | 27/27; 0/0; 0.18 | 27/27; 0/0; 0.18 | 12/27; 11/0; 54.58 |
| In-bank | 1 | 27/27; 0/0; 0.18 | 27/27; 0/0; 0.18 | 13/27; 11/0; 51.04 |
| In-bank | 2 | 27/27; 0/0; 0.18 | 27/27; 0/0; 0.18 | 13/27; 11/0; 51.05 |
| Out-of-bank | 1 | 27/27; 0/0; 0.18 | 27/27; 0/0; 0.18 | 12/27; 11/0; 54.58 |
| Out-of-bank | 2 | 27/27; 0/0; 0.18 | 27/27; 0/0; 0.18 | 12/27; 11/0; 54.59 |

computation latency. Bad-data labels come from the inherited window classifier. Neither missingness nor corruption proves malicious intent. The composition preserves physical-source and affected-PMU identities but does not run a joint physical/cyber Bayesian filter; the physical bank is unavailable on the retrospective replay of the competition reference record under its startup noise-scale contract.

**TABLE 8.** **Persistent-line coverage and stopping outcomes across six repeated strata, including both fixed ExtraTrees comparators. Normal controls are the same controls used in the continuous table, not additional exposure. Chen *et al.* normal exposure ends at its first stop; the other methods monitor the full available record.**

| Method | Unavailable | Correct first | Pre-change | Censored | Normal incidence | Exposure (h) |
|---|---|---|---|---|---|---|
| Physical 5-model | 0 | 162/162 | 0 | 0 | 0/72 | 11.760 |
| Physical nominal | 0 | 162/162 | 0 | 0 | 0/72 | 11.760 |
| ExtraTrees | 0 | 162/162 | 0 | 0 | 0/72 | 11.760 |
| ET short | 0 | 162/162 | 0 | 0 | 0/72 | 11.760 |
| Chen *et al.* | 0 | 74/162 | 66 | 0 | 72/72 | 0.701 |

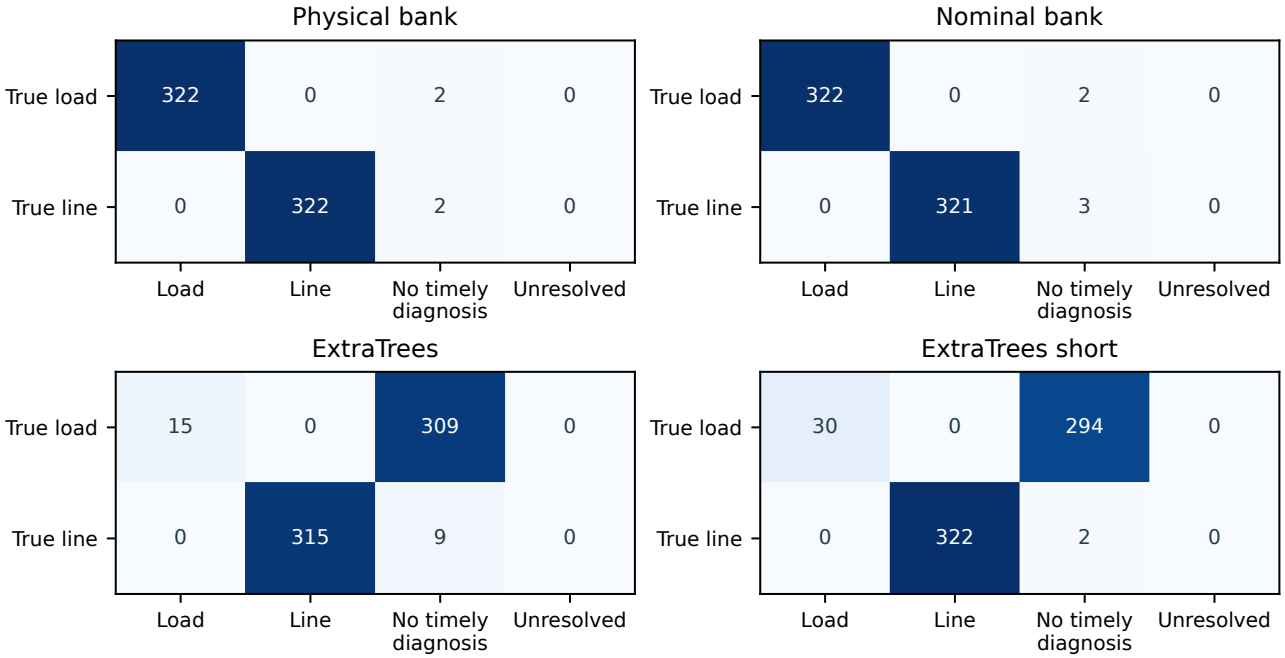



**FIGURE 1.** **Posthoc confusion audit of the fixed continuous test, with 324 realizations per true family. No timely diagnosis includes omitted and late diagnoses. Counts reuse the registered episode matches; all 432 prediction JSON files used were checked against the stored prediction hashes. These panels do not include unmatched alarms or constitute a new predeclared endpoint.**

**TABLE 9.** **Timely decisions by source in the continuous test. These are correlated realizations of the registered physical schedules, not source-excluded generalization.**

| Family / asset | Physical bank | Nominal bank | ExtraTrees | ET short |
|---|---|---|---|---|
| Line / LINE14-15 | 36/36 | 36/36 | 27/36 | 34/36 |
| Line / LINE16-17 | 36/36 | 36/36 | 36/36 | 36/36 |
| Line / LINE17-18 | 48/48 | 48/48 | 48/48 | 48/48 |
| Line / LINE23-24 | 30/30 | 30/30 | 30/30 | 30/30 |
| Line / LINE26-27 | 24/24 | 24/24 | 24/24 | 24/24 |
| Line / LINE26-29 | 24/24 | 23/24 | 24/24 | 24/24 |
| Line / LINE3-4 | 42/42 | 40/42 | 42/42 | 42/42 |
| Line / LINE4-14 | 46/48 | 48/48 | 48/48 | 48/48 |
| Line / LINE9-39 | 36/36 | 36/36 | 36/36 | 36/36 |
| Load / BUS15 | 42/42 | 42/42 | 6/42 | 6/42 |
| Load / BUS16 | 42/42 | 42/42 | 0/42 | 3/42 |
| Load / BUS20 | 12/12 | 12/12 | 6/12 | 12/12 |
| Load / BUS24 | 42/42 | 42/42 | 0/42 | 0/42 |
| Load / BUS26 | 18/18 | 18/18 | 0/18 | 0/18 |
| Load / BUS3 | 42/42 | 42/42 | 3/42 | 3/42 |
| Load / BUS31 | 40/42 | 40/42 | 0/42 | 0/42 |
| Load / BUS4 | 36/36 | 36/36 | 0/36 | 6/36 |
| Load / BUS7 | 48/48 | 48/48 | 0/48 | 0/48 |

**TABLE 10. Paired physical-bank minus learned-comparator differences. Timely attribution is in percentage points; extras are per monitored event-record hour, excluding 12 s of startup from each 120-s record. The registered rate differences and 95-percent parent-bootstrap bounds are exactly rescaled by 120/108; original counts, matches and intervals remain stored.**

| Network | Noise | Method | Reference | Timely difference | Extra-rate difference |
|---|---|---|---|---|---|
| Matched | 1 | Physical bank | ExtraTrees short | 44.44 [40.48, 47.92] | -7.41 [-16.67, 0.00] |
| Matched | 1 | Physical bank | ExtraTrees | 48.15 [44.44, 51.67] | -9.26 [-20.37, 0.00] |
| Matched | 1 | Nominal bank | ExtraTrees short | 44.44 [40.74, 48.04] | -7.41 [-14.81, 0.00] |
| Matched | 1 | Nominal bank | ExtraTrees | 48.15 [44.44, 51.67] | -9.26 [-20.37, 0.00] |
| Matched | 2 | Physical bank | ExtraTrees short | 46.30 [42.75, 49.12] | 0.00 [0.00, 0.00] |
| Matched | 2 | Physical bank | ExtraTrees | 50.00 [46.30, 53.70] | -7.41 [-14.81, -1.85] |
| Matched | 2 | Nominal bank | ExtraTrees short | 46.30 [42.86, 49.12] | 0.00 [0.00, 0.00] |
| Matched | 2 | Nominal bank | ExtraTrees | 50.00 [46.38, 53.70] | -7.41 [-14.81, -1.85] |
| In-bank | 1 | Physical bank | ExtraTrees short | 44.44 [40.62, 47.92] | -5.56 [-14.81, 5.56] |
| In-bank | 1 | Physical bank | ExtraTrees | 48.15 [44.44, 51.67] | -5.56 [-14.81, 5.56] |
| In-bank | 1 | Nominal bank | ExtraTrees short | 44.44 [40.62, 47.92] | -3.70 [-12.96, 7.41] |
| In-bank | 1 | Nominal bank | ExtraTrees | 48.15 [44.44, 51.67] | -3.70 [-12.96, 5.56] |
| In-bank | 2 | Physical bank | ExtraTrees short | 47.22 [43.18, 50.98] | 1.85 [0.00, 5.56] |
| In-bank | 2 | Physical bank | ExtraTrees | 50.00 [46.49, 53.70] | -5.56 [-11.11, 0.00] |
| In-bank | 2 | Nominal bank | ExtraTrees short | 47.22 [43.33, 51.04] | 3.70 [0.00, 9.26] |
| In-bank | 2 | Nominal bank | ExtraTrees | 50.00 [46.30, 53.70] | -3.70 [-11.11, 3.70] |
| Out-of-bank | 1 | Physical bank | ExtraTrees short | 42.59 [38.60, 46.43] | 59.26 [40.74, 79.63] |
| Out-of-bank | 1 | Physical bank | ExtraTrees | 46.30 [42.11, 50.00] | 61.11 [40.74, 81.48] |
| Out-of-bank | 1 | Nominal bank | ExtraTrees short | 41.67 [36.11, 46.67] | 75.93 [51.85, 100.00] |
| Out-of-bank | 1 | Nominal bank | ExtraTrees | 45.37 [41.11, 50.00] | 77.78 [50.00, 105.56] |
| Out-of-bank | 2 | Physical bank | ExtraTrees short | 45.37 [41.11, 50.00] | 55.56 [38.89, 72.22] |
| Out-of-bank | 2 | Physical bank | ExtraTrees | 48.15 [43.59, 52.50] | 46.30 [27.78, 66.67] |
| Out-of-bank | 2 | Nominal bank | ExtraTrees short | 45.37 [41.11, 50.00] | 72.22 [46.30, 98.15] |
| Out-of-bank | 2 | Nominal bank | ExtraTrees | 48.15 [43.75, 52.56] | 62.96 [35.19, 88.89] |

**TABLE 11. Descriptive persistence audit after first detection on sustained line openings. All four procedures first identify 162/162 lines. Extras are later unmatched alarms; unmatched OFF counts only restoration messages on unmatched episodes. Open-at-end messages are expected for sustained events and are not automatically errors. False-negative reports include detection delay and later inactive labels. Chen is a first-stop method and is excluded from this recurrent-monitoring diagnostic.**

| Method | Openings | Extras | Unmatched OFF | Open at end | FN reports |
|---|---|---|---|---|---|
| Physical bank | 162 | 0 | 0 | 162 | 810 |
| Nominal bank | 162 | 0 | 0 | 162 | 810 |
| ExtraTrees | 162 | 164 | 164 | 16 | 335374 |
| ExtraTrees short | 162 | 179 | 178 | 61 | 182600 |

**TABLE 12. Retrospective measurement-integrity components over 180 windows. Hits require family and PMU identity. Missingness is observed availability; corrupted-data identification is inherited from the window classifier. These are not counts of attributed cyberattacks.**

| Component | Positive-window hits | FP | FN |
|---|---|---|---|
| Missing | 14/14 | 0 | 0 |
| Corrupt | 1/2 | 2 | 1 |

## APPENDIX H CONTROLLED COMPONENT ABLATION

The fixed protocol and all output tables are part of the reproducibility package. Each of four arms accounts for all 24 intended records (18 continuous, three normal and three persistent): the evaluation audit records 96/96 available outputs and zero failures. The trial is posthoc and no score threshold is calibrated to these outcomes. Each normal control is a 600-s record with 588 s of evaluated exposure after startup, for 1,764 s total.

*a: Exposure convention for the ablation alarm rate.*

The ablation table reports extras per event-hour with the complete 120-s event record, including startup, as denominator. Its continuous exposure is $18 \times 120 = 2160$ s. The main article's confirmatory alarm-rate figure instead uses monitored exposure after the 12-s startup. Under that convention the ablation exposure would be $18 \times 108 = 1944$ s and each reported rate difference and interval bound would be multiplied by $120/108$. Counts, event matches and the stored table are unchanged; these two rate conventions must not be compared without this conversion.

## APPENDIX I EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.

For a load ON alternative, write the innovation as $e_k - as_k$. The affine conditional state is $m_k + ad_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{21}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{22}$$

$$C = -\frac{1}{2} \sum_k \left( e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{23}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a}U - \tfrac{1}{2}\widehat{a}^2 G - C_{ref}. \tag{24}$$

Here $C_{ref}$ is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c)P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The development comparison contains no such post-startup missing observations.

**TABLE 13. All outcomes, including restoration messages on unmatched episodes and episodes open at the record end.**

| method | lane | available | events | timely | extras | false restorations | open eof |
|---|---|---|---|---|---|---|---|
| Marginal + carry | continuous | 18 | 108 | 108 | 1 | 1 | 0 |
| Marginal + carry | normal | 3 | 0 | 0 | 0 | 0 | 0 |
| Marginal + carry | persistent line | 3 | 3 | 3 | 0 | 0 | 3 |
| Marginal + reset | continuous | 18 | 108 | 75 | 166 | 130 | 11 |
| Marginal + reset | normal | 3 | 0 | 0 | 0 | 0 | 0 |
| Marginal + reset | persistent line | 3 | 3 | 3 | 0 | 0 | 3 |
| Profile + carry | continuous | 18 | 108 | 108 | 1 | 1 | 0 |
| Profile + carry | normal | 3 | 0 | 0 | 0 | 0 | 0 |
| Profile + carry | persistent line | 3 | 3 | 3 | 0 | 0 | 3 |
| Profile + reset | continuous | 18 | 108 | 75 | 166 | 130 | 11 |
| Profile + reset | normal | 3 | 0 | 0 | 0 | 0 | 0 |
| Profile + reset | persistent line | 3 | 3 | 3 | 0 | 0 | 3 |

**TABLE 14. Paired descriptive differences with 95-percent physical-parent bootstrap intervals; no multiplicity-adjusted superiority claim.**

| method | reference | metric | difference | lower95 | upper95 |
|---|---|---|---|---|---|
| Marginal + carry | Profile + carry | timely fraction | 0.000 | 0.000 | 0.000 |
| Marginal + carry | Profile + carry | extras per event-hour | 0.000 | 0.000 | 0.000 |
| Marginal + carry | Marginal + reset | timely fraction | 0.306 | 0.241 | 0.380 |
| Marginal + carry | Marginal + reset | extras per event-hour | -275.000 | -325.000 | -223.333 |
| Marginal + reset | Profile + reset | timely fraction | 0.000 | 0.000 | 0.000 |
| Marginal + reset | Profile + reset | extras per event-hour | 0.000 | 0.000 | 0.000 |
| Profile + carry | Profile + reset | timely fraction | 0.306 | 0.241 | 0.380 |
| Profile + carry | Profile + reset | extras per event-hour | -275.000 | -325.000 | -223.333 |

**TABLE 15. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.**

| Procedure | Timely | Extras | Early OFF | No OFF |
|---|---|---|---|---|
| Single state, short GLR | 37/54 | 37 | 0 | 4 |
| Nominal history bank | 44/54 | 6 | 0 | 10 |
| Five-model history bank | 51/54 | 3 | 0 | 3 |

**TABLE 16. What the causal component resolves in the second development campaign. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.**

| Method | Load timely | Line timely | Extras | Normal alarms |
|---|---|---|---|---|
| Physical (5 models) | 26/27 | 27/27 | 4 | 0 |
| Physical nominal | 27/27 | 27/27 | 7 | 0 |
| ExtraTrees ref. | 0/27 | 27/27 | 3 | 0 |
| ET short | 0/27 | 27/27 | 3 | 0 |
| Onset witness | 25/27 | 27/27 | 0 | 0 |

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

## APPENDIX J DEVELOPMENT STREAMS AND ADMISSION CONTROL

### A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

In the second development campaign, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 16). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the fixed adapters; it neither rules out supervised load detection nor reproduces the historical multi-class hackathon score.

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

### B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 3 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing

failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and the mass-increment identity in the main article separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

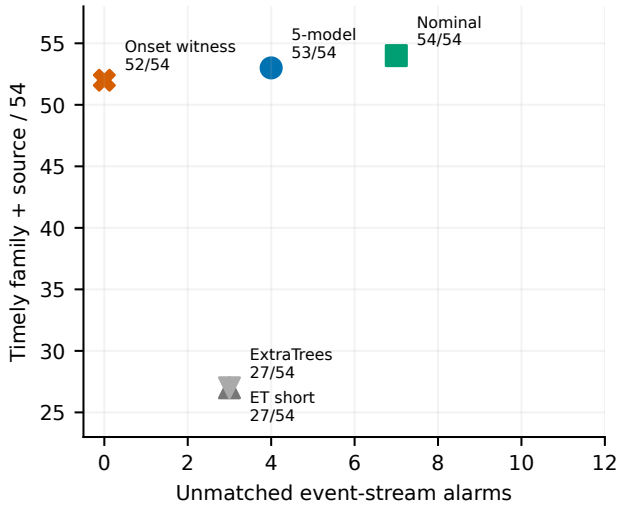



**FIGURE 2. The decision that an accuracy-only table hides. Development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.**

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative result left the five-model and nominal banks unchanged for the independent evaluation.

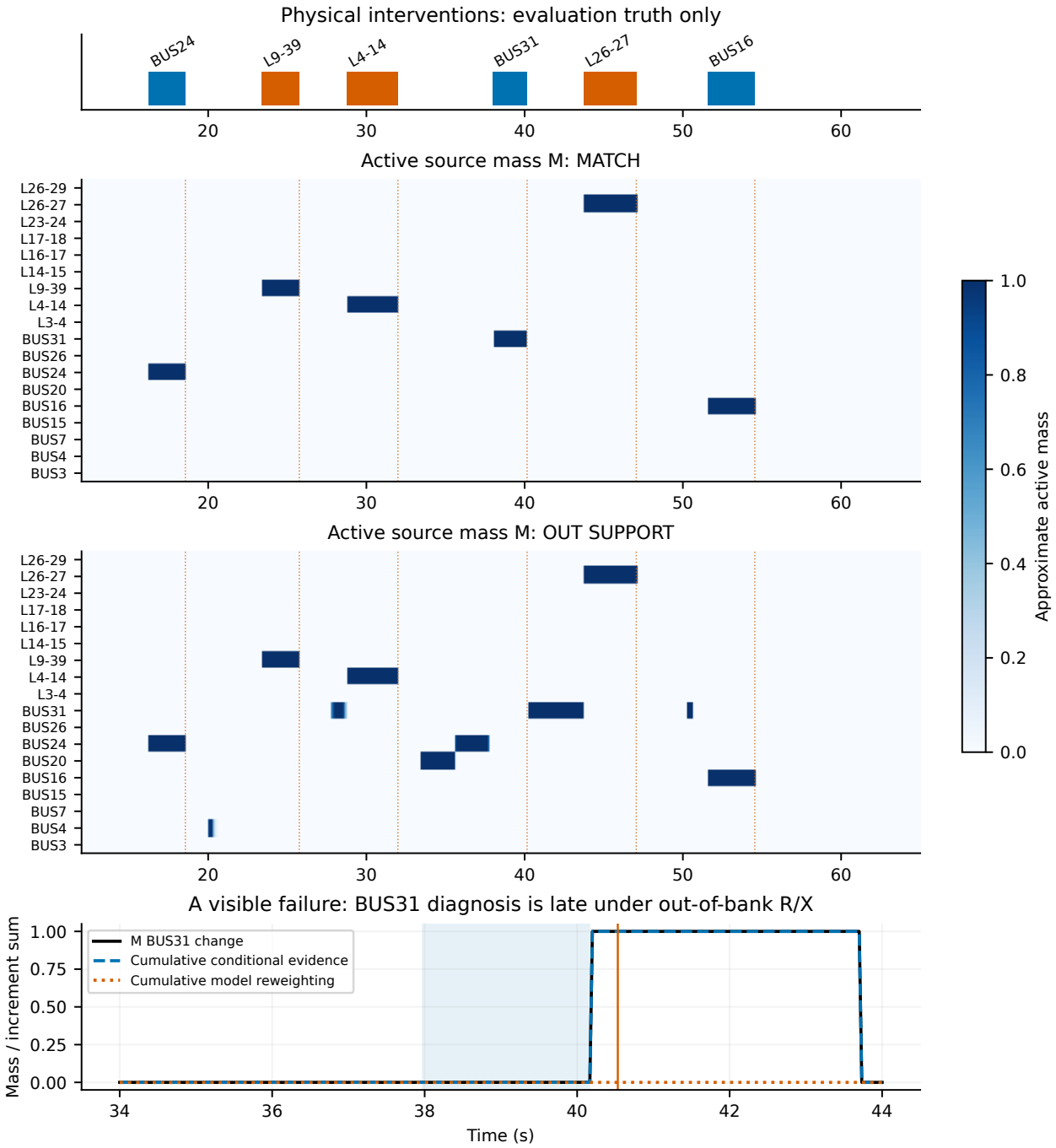



**FIGURE 3.** **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in one development parent. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

## APPENDIX K DETECTABILITY COMPUTATION AND NOISE-STRESS PROTOCOL

*a: Operators.*

The information analysis of the main article uses the nominal bank built by the evaluated implementation for one held-out calibration record of a later campaign whose runtime is byte-identical to the independent evaluation runtime. The independent-evaluation banks had been evicted by the storage-retention policy; the nominal operators differ between records only through the prefix-estimated operating factors, which remain within $2 \times 10^{-5}$ of unity.

*b: Signatures.*

For each source $i$, the noise-free switched predictor is propagated from a zero pre-event deviation with the ON operator at half-phase 0 followed by the stay operator (loads at severity 0.1). Each candidate filter $j \in \{0, \dots, 18\}$ is run on these reports with the quiet-regime gains $K_o$ and inverse innovation covariances $S_o^{-1}$ of the implementation. The cumulative sums $\sum e^\top S^{-1} e$, $\sum s^\top S^{-1} e$ and $\sum s^\top S^{-1} s$ give the signature information of the main article with the competitor's severity profiled on $[-0.2, 0.2]$. As a check, the true-history filter at the true severity leaves a residual energy below $10^{-8}$. The detection threshold uses the birth hazard $1 - \exp[-(1/60)/30]$ shared equally by the 18 cells.

*c: Reset cost.*

For every intervention of the controlled study, the noise-free truth is propagated through ON, the active interval, OFF and the gap until the next onset. At the inferred closure time of the carry arm, the first 114 coordinates of the true augmented state are zeroed, and the no-intervention and all birth filters are run on the remaining truth in the recovery covariance regime. The largest profiled gain of a birth over the no-intervention history is compared with the alarm evidence $\lambda^\star$. Harm in the reset arm is an unmatched alarm between the OFF of the intervention and the next onset, or a lost timely attribution of the next intervention.

*d: Noise stress.*

Observations are regenerated from the stored deterministic trajectories with the simulator's own noise stream and seed: $\varepsilon = c + \alpha(v - c)$ for multiplier $\alpha$, where $v$ is the stream and $c$ its normalization center, exactly as the campaign exporter. The $\alpha = 1$ observations reproduce the original files to $6 \times 10^{-10}$ and the inference reproduces all 18 published messages of the parity record. The TVE scenarios add independent zero-mean Gaussian phasor noise per PMU and report, with relative magnitude and angle (radian) standard deviations of $5 \times 10^{-4}$ or $10^{-3}$; the added voltage-angle noise is passed through the simulator's causal 21-report quadratic frequency and ROCOF estimator. The nominal bank is rebuilt from the stored derivative matrices with the noise scale estimated from the noisy prefix. The operating point remains at its $\alpha = 1$ estimate because derivatives exist only there. Estimated angle noise scales of the TVE scenarios are about 48 times the calibrated scale, outside the implemented support $[0.1, 10]$; for this test only the support was widened to $[0.1, 1000]$. Two jobs ran concurrently with another campaign on the same desktop CPU, so the reported processing rates are conservative.

*e: Reproduction.*

Every number and figure of the analysis is regenerated from the stored operators and outcomes by the reproducibility package, which also contains the noise-stress generator.

## APPENDIX L OPERATING-POINT SIGNATURE EXPERIMENT

### A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of the stream method.

Let $\eta$ denote a scalar operating change and $a$ the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h}, \qquad E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{25}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed $Q_i^0$ is the coefficient of $a^2$, obtained from the centered second difference divided by $2h^2$. After removing the fixed common angle drift, the three predicted histories are

$$\begin{aligned} \mu_i^{\mathrm{N}}(a) &= y_0 + aD_i^0 + a^2 Q_i^0, \\ \mu_i^{\mathrm{R}}(a) &= \mu_i^{\mathrm{N}}(a) + \widehat{b}, \\ \mu_i^{\mathrm{E}}(a) &= \mu_i^{\mathrm{R}}(a) + a\widehat{\eta} E_i. \end{aligned} \tag{26}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

### B. TEST CASES AND DECISION CONTRACT

The research plant supplies all trajectories. The scalar $\eta$ scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction

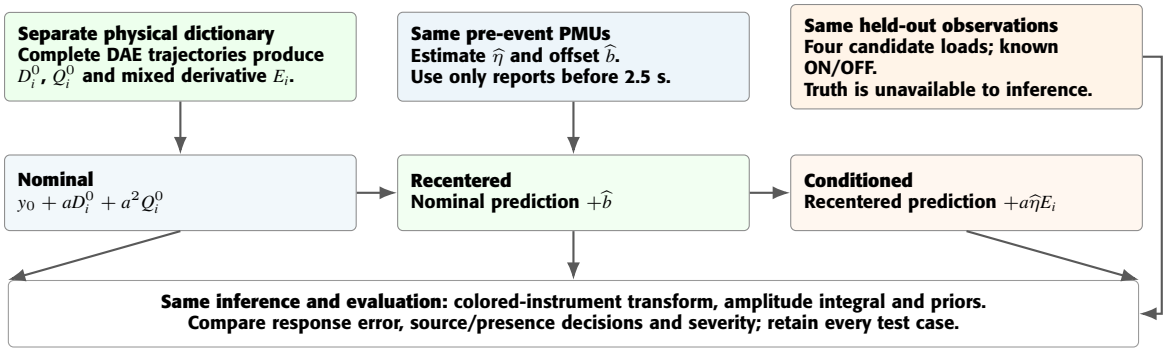



**FIGURE 4.** Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.

uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three fixed instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its stored instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

### C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 5 retains that case rather than reporting only shifted operation.

The derivative stability checks change $E_i$ by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the fixed impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the fixed artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

## APPENDIX M RECOVERY GEOMETRY AND SOURCE SEPARATION

### A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new intervention begins. For the exact observation map $\Psi$ and the same pre-intervention history $h$, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{27}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

**TABLE 17. Controlled signature test, separate from causal detection. Response error is the median whitened incremental error at the generating source/severity over 80 physical cases. Joint correctness includes presence and source on 240 noisy event records; source-only ranking conditions on a load. Severity error is median absolute relative error at the generating source. Normal selections use 15 fixed windows, not normal-hours exposure.**

| Model | Response error (%) | Joint correct | Source-only correct | Normal event selections | Severity error (%) |
|---|---|---|---|---|---|
| Nominal | 1.25 | 114/240 | 114/240 | 12/15 | 3.45 |
| Recentered | 1.25 | 240/240 | 240/240 | 0/15 | 1.2 |
| Conditioned | 0.0561 | 240/240 | 240/240 | 0/15 | 0.329 |

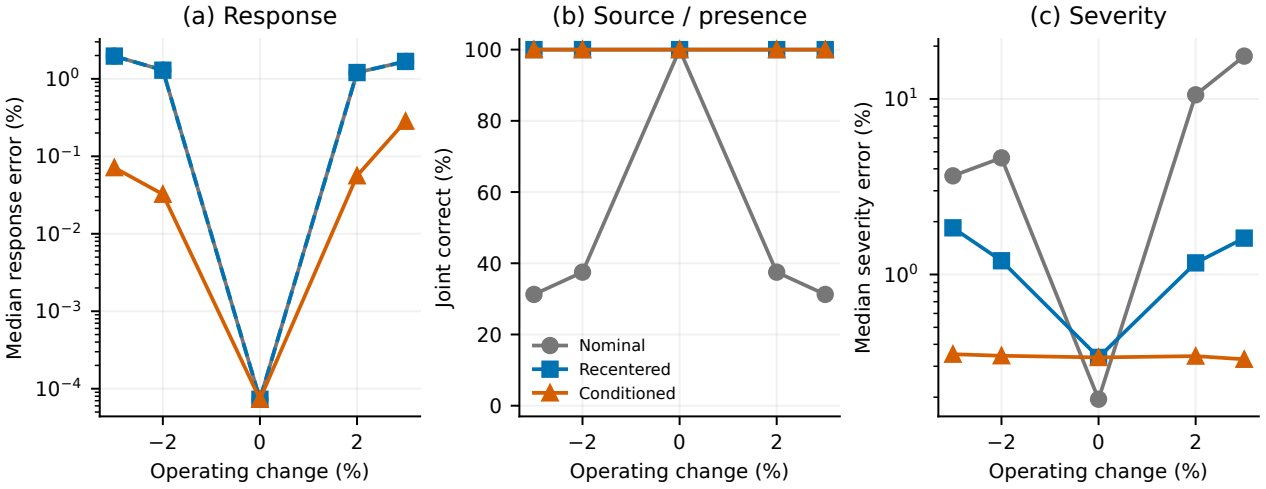



**FIGURE 5. Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.**

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response $s_h$ at specified severity and context, the distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{28}$$

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [2], [3].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The separately compiled Historical Technical Notes retain the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

### B. DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete nonlinear trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied.

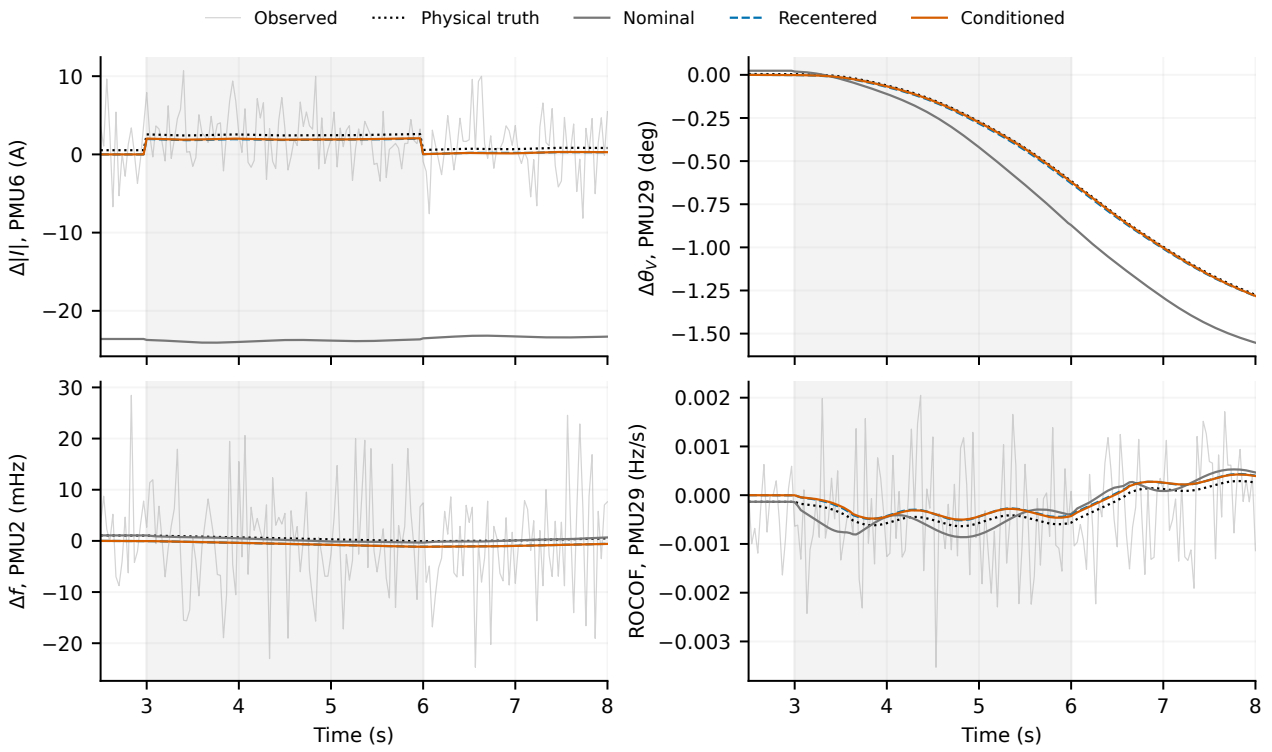



**FIGURE 6.** **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate measurement operator is followed by its fixed innovations transform. For a geometric control, we profile nonnegative competing amplitudes. Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios exceeds one (Fig. 7). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an improvement of the stream method nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

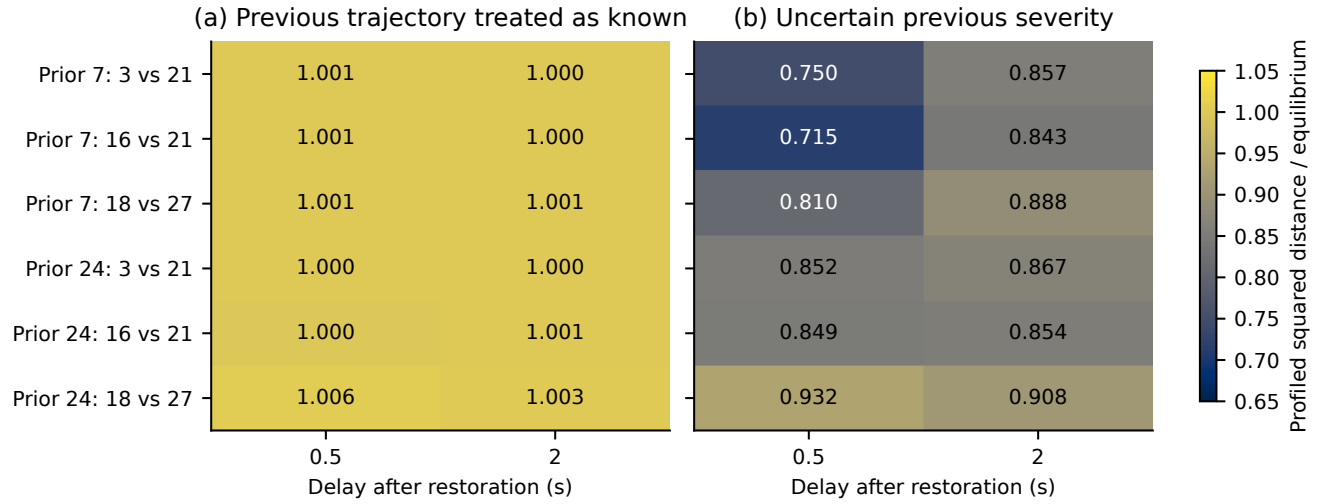



**FIGURE 7.** **Full nonlinear physical control, not a performance comparison of the stream method. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

## REFERENCES

[1] S. Särkkä, *Bayesian Filtering and Smoothing*. Cambridge, U.K.: Cambridge University Press, 2013.

[2] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.

[3] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.